

Concept bottleneck models

Making neural networks predict through human-understandable concepts

Images, time series and text

Input → Named concepts → Prediction

What did the model detect?

What happens when we correct it?

ES 667 · Deep learning

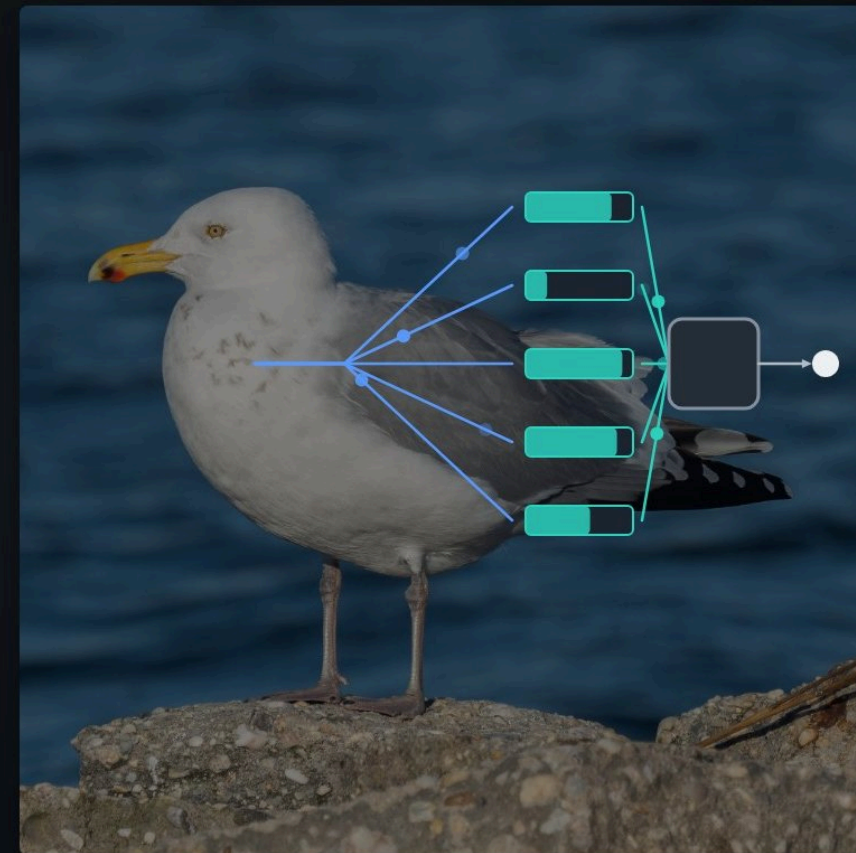
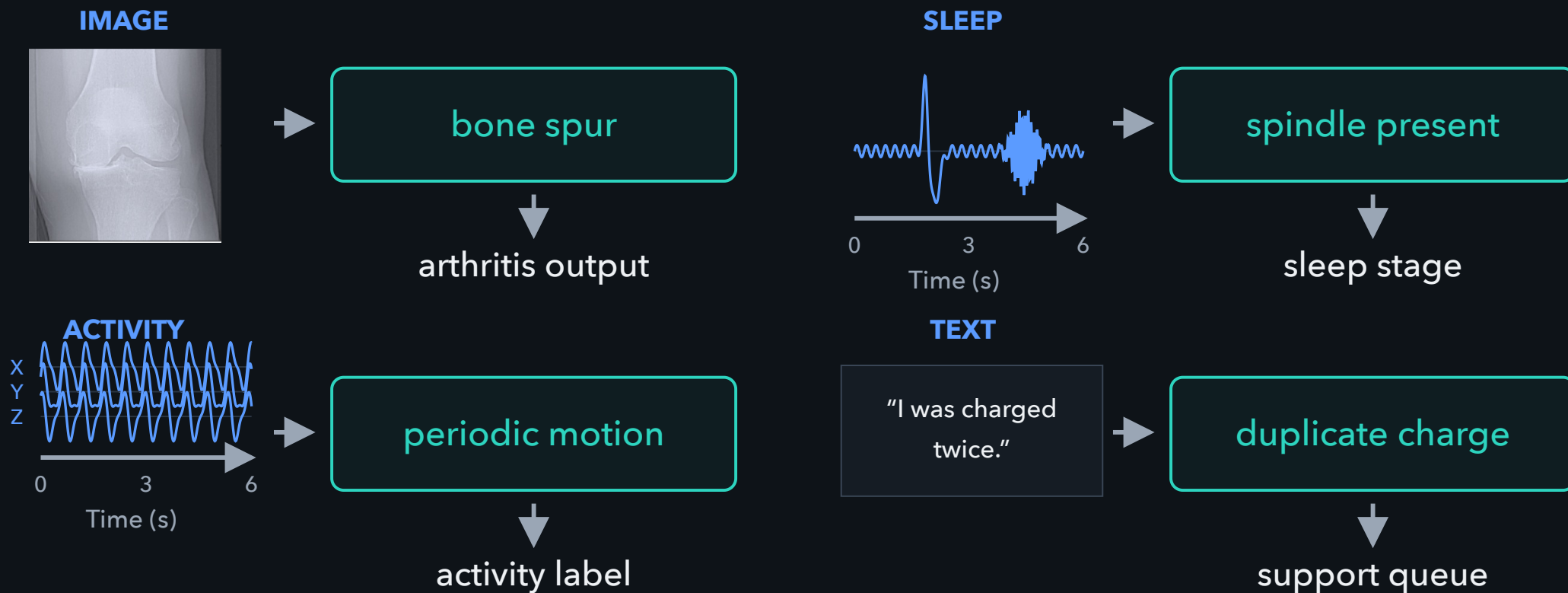


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

Concepts can describe images, signals or text



The input changes. The task still predicts through named variables.

An image example: correcting a bone-spur belief



ONE NAMED BELIEF

Bone spur

0.18 → 1.00

Radiologist supplies a correction

ILLUSTRATIVE TASK HEAD

72.2% → 93.1%

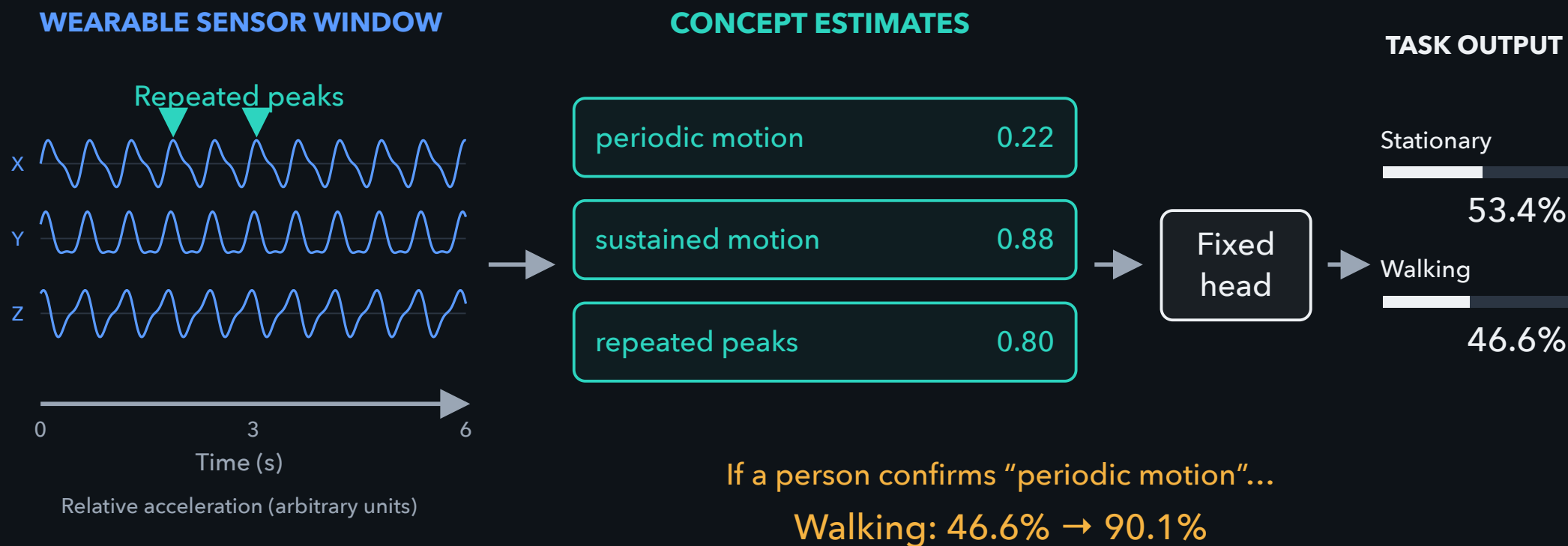
Illustrative severe-class share

Same image. Same weights.

Only the concept value changes.

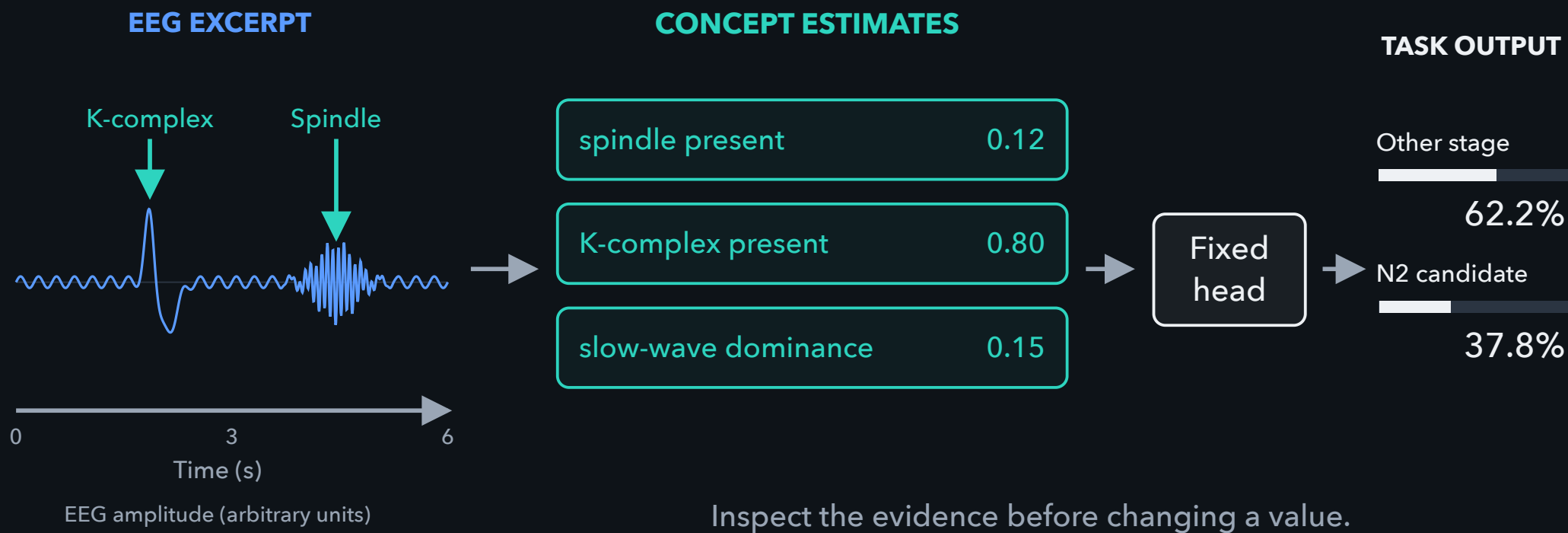
The image stays fixed. Correcting one concept changes the severity score.

Activity recognition: a repeated movement



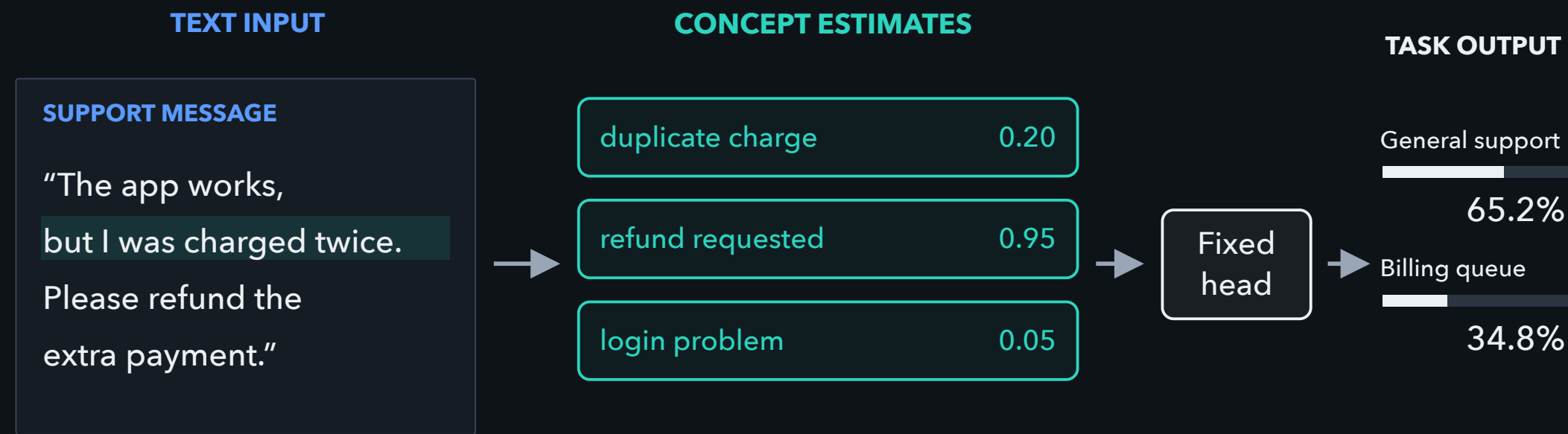
A person can check the repeated pattern, then correct "periodic motion."

Sleep staging: a burst in the EEG



A concept can refer to an event within a time window.

Text classification: "I was charged twice"



"Charged twice" is the evidence.

Inspect the evidence before changing a value.

The concept is the meaning of a claim, not just a highlighted word.

Interpretability

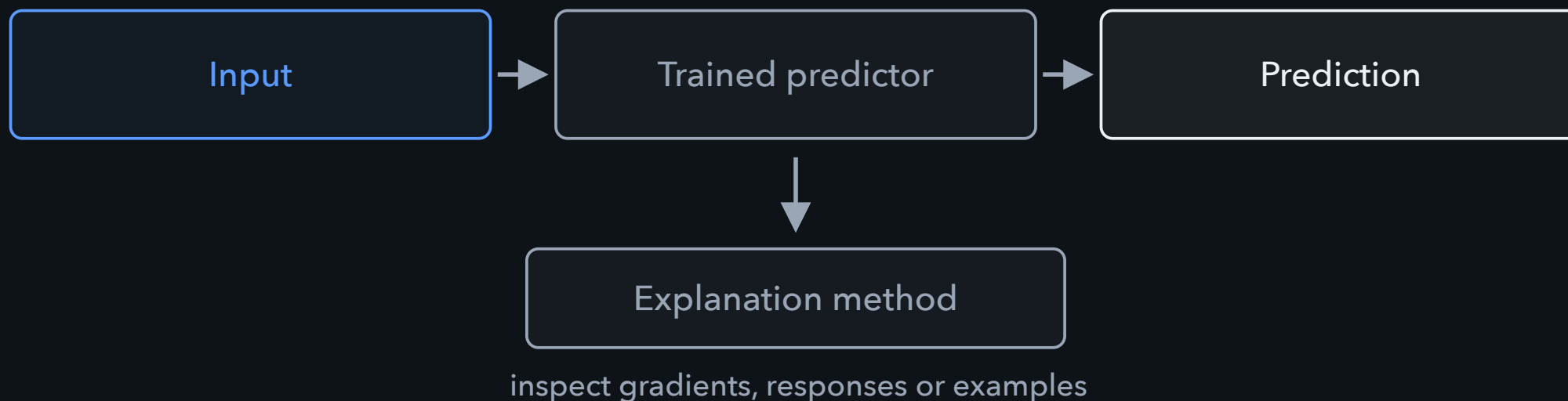


Inspect a trained predictor, or design an editable one.

What can an explanation tell us?

Post-hoc explanations inspect a trained predictor

PREDICT FIRST



Attribution

which pixels / times / words?

Perturbation

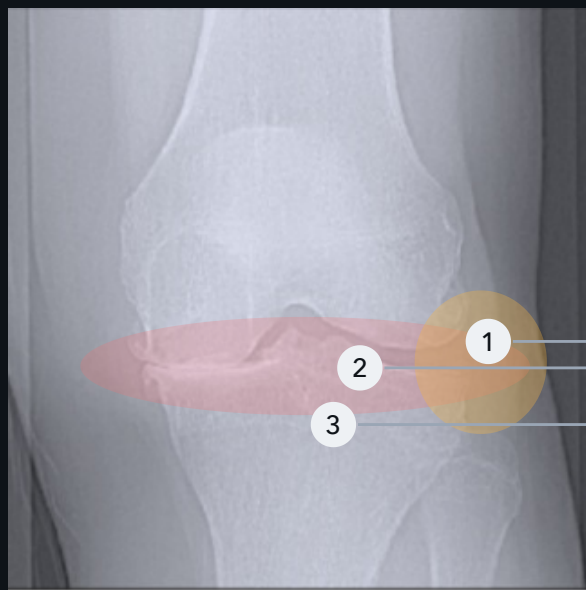
what changes if input is masked?

Concept probe

sensitivity to a named concept?

Attribution, input perturbations and concept probes ask different questions about the predictor.

A heatmap marks a region. What could that region contain?



Educational knee image

Bone edge

Could there be a bone spur?

Joint gap

Could the joint space be narrow?

Bone below the joint

Could there be sclerosis?

ILLUSTRATIVE BELIEFS

bone spur

.18

joint narrowing

.91

sclerosis

.35

The overlay itself does not assign these concepts.

The labels name candidate findings to inspect. The heatmap does not verify them.

A real black box: CLIP looks at a gull

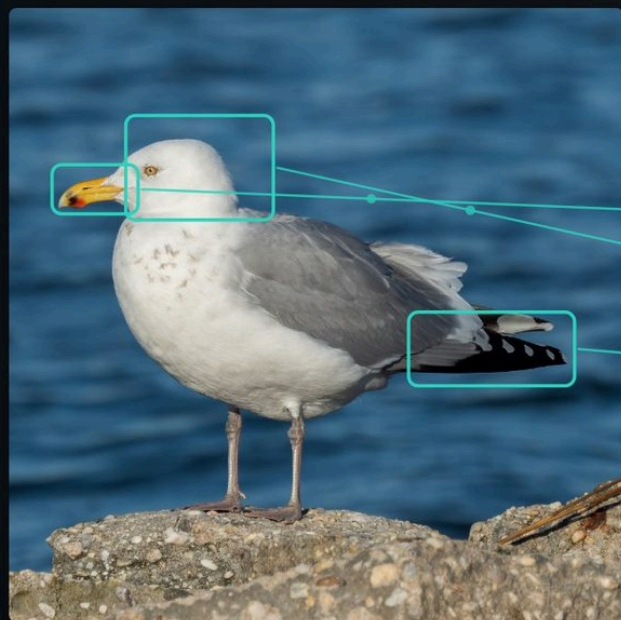


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

named properties

bill shape



head colour



wingtip colour

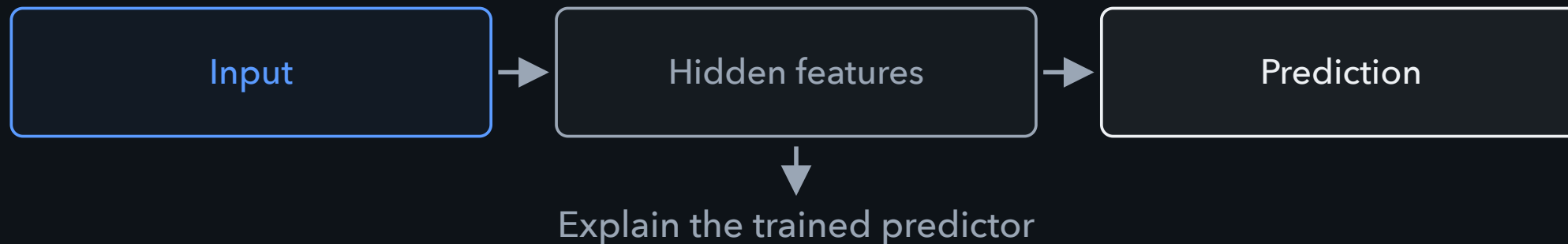


each claim can be checked

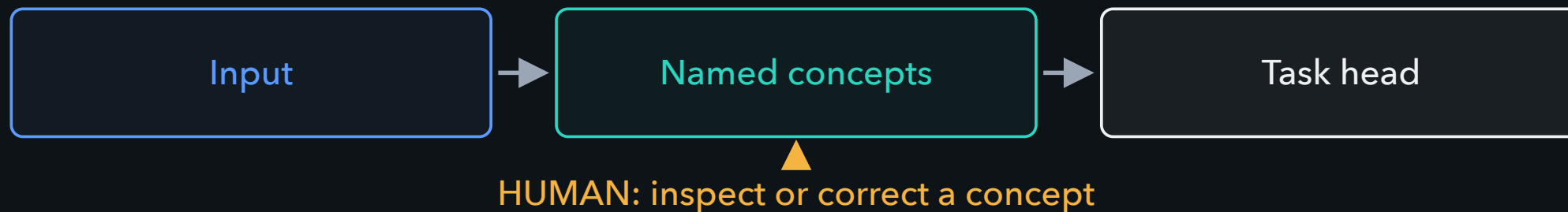
A computed heatmap shows where the score is sensitive, not which property was detected.

Interpretability by design puts concepts on the path

POST-HOC

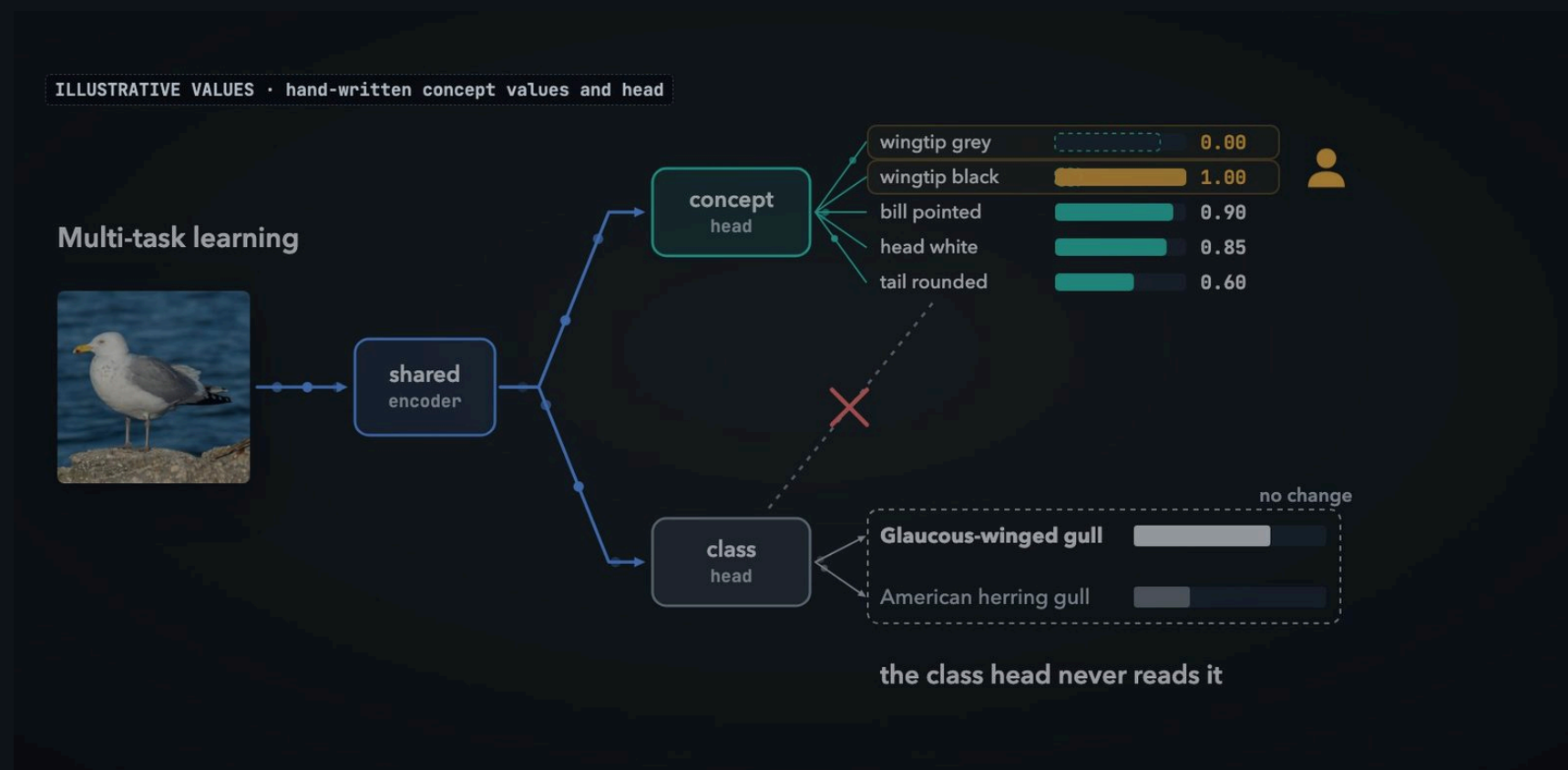


BY DESIGN (INTRINSIC / ANTE-HOC)



A strict CBM makes the named concept vector the task head's only input.

The same gull through a concept bottleneck

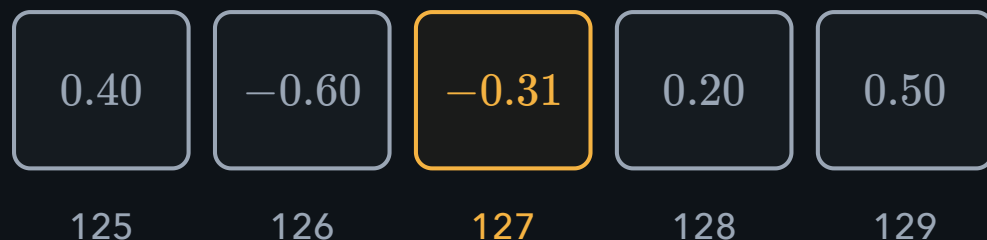


Concepts on the path can be edited; concepts beside the path cannot.

A useful edit needs a named coordinate

UNNAMED HIDDEN COORDINATES

Each number has an index.

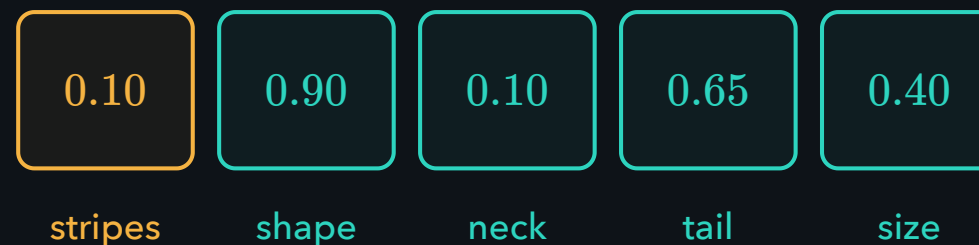


$$h_{127} : -0.31 \longrightarrow 0.70$$

What would this change mean?

DECLARED CONCEPT COORDINATES

Each number has a definition.

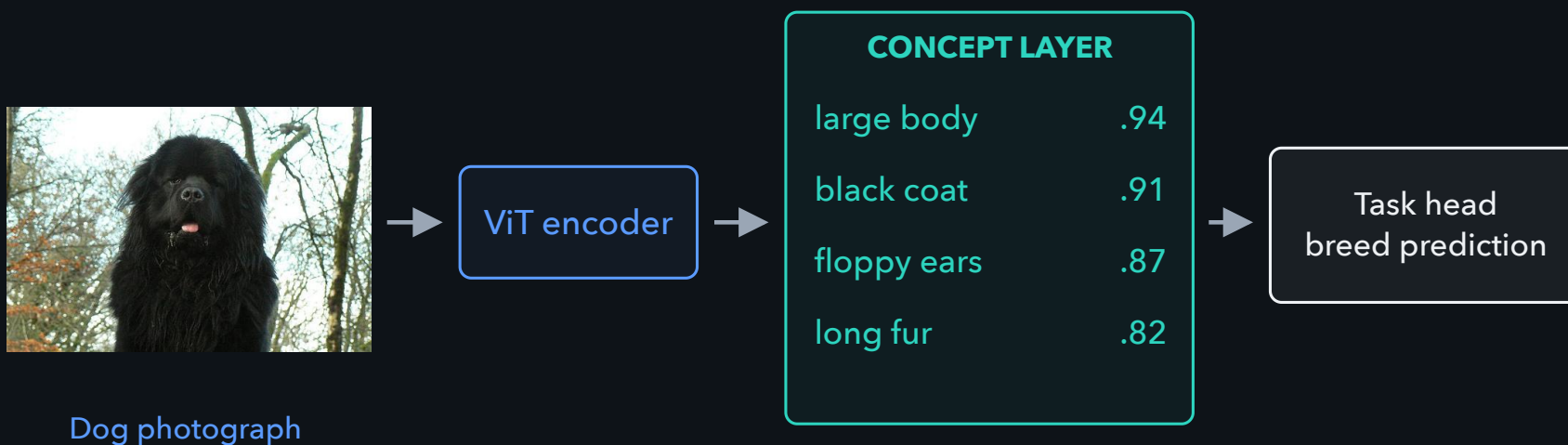


$$\hat{c}_{\text{stripes}} : 0.10 \longrightarrow 1.00$$

A human can check the stripes.

“Stripes present” gives the human a specific claim to correct.

A ViT can predict concepts before a dog breed

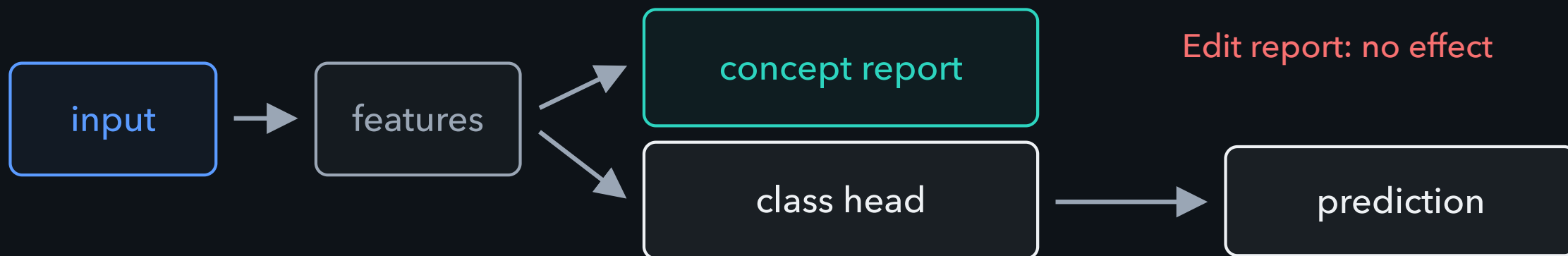


$$\hat{\mathbf{c}} = g_{\theta}(x) \quad \hat{y} = f_{\phi}(\hat{\mathbf{c}})$$

The encoder learns image features. The breed head receives the four concept estimates.

Multi-task learning can leave concepts beside the prediction

MULTI-TASK: CONCEPTS CAN BE A SIDE REPORT



STRICT CBM: THE TASK HEAD MUST USE CONCEPTS



Trace the arrows: which class head actually reads the editable value?

Binary, categorical and continuous concepts

Binary



Bone spur present?

$$c_j \in \{0, 1\}$$

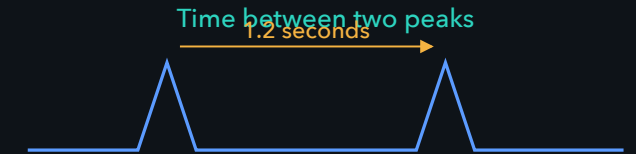
Categorical



Wing colour

grey · black · other

Continuous



Mean peak interval

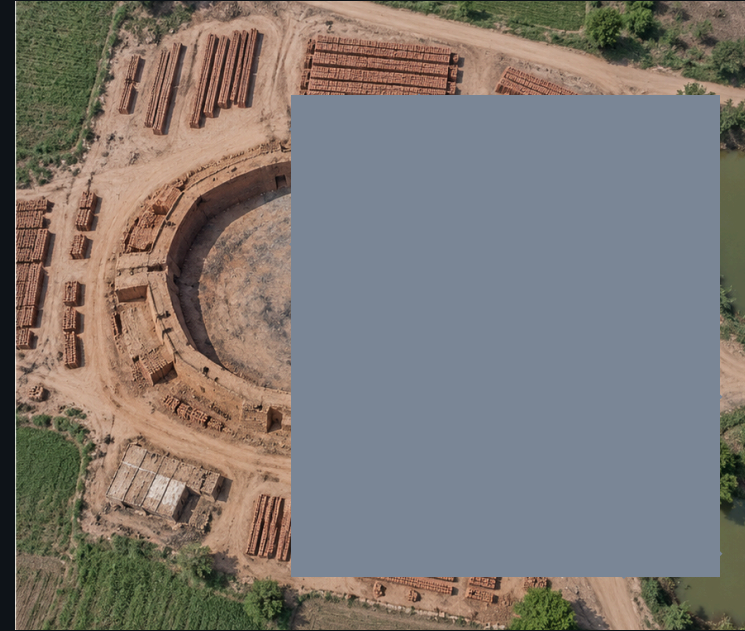
Defined in seconds

Declare the type and scale. A soft estimate also needs calibration.

Absent and unobservable need different labels



ABSENT: clear view



UNKNOWN: area masked

Use an unknown value when the relevant evidence cannot be seen.

The concept value also needs a meaning

Discrete

$$c_j \in \{0, 1\}$$

A recorded category: absent or present.

Continuous

$$\hat{c}_j = 0.72$$

A soft score. Calibration needs testing.

Distributional

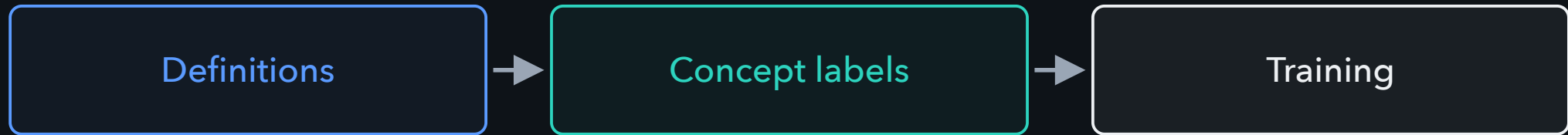
$$P(C_j = 0, 1 \mid x) = (0.28, 0.72)$$

A distribution over possible values.

Unknown visibility is different from known absence.

A scalar probability can parameterize a binary distribution. Richer distributions model more.

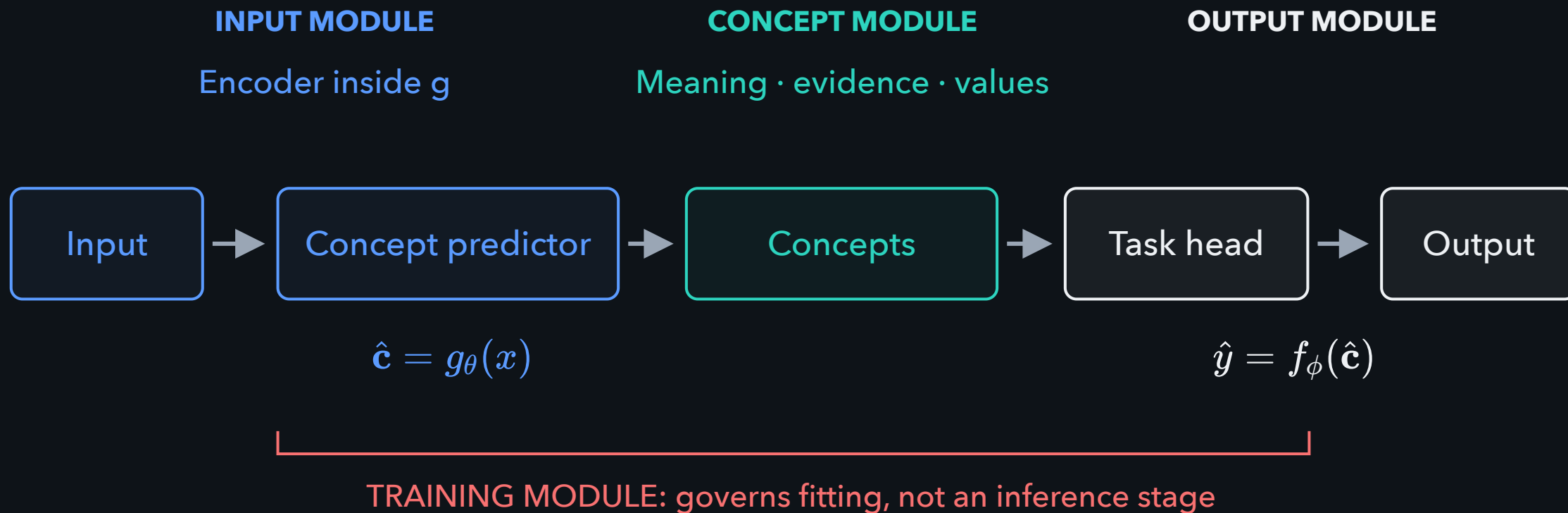
Building a CBM



Concept labels connect a definition to training data.

Where do the concepts and training signals come from?

Four choices shape a concept bottleneck



Choose the input representation, concept interface, predictor and training procedure.

A concept needs both a meaning and a detector

SEMANTICS: WHAT DOES IT MEAN?

Concept: duplicate charge

The customer reports two payments for the same purchase.

The definition follows the claim, including negation and uncertainty.

GROUNDING: WHAT IS THE EVIDENCE?

"I was charged twice."



duplicate charge = present

A label, span, detector or pretrained model can supply a grounding signal.

Choosing a good name and learning a reliable detector are separate jobs.

A clear definition does not guarantee that the detector follows it.

One concept: meaning, evidence and representation



Generated teaching scene

SEMANTICS

An identifiable chimney structure

GROUNDING

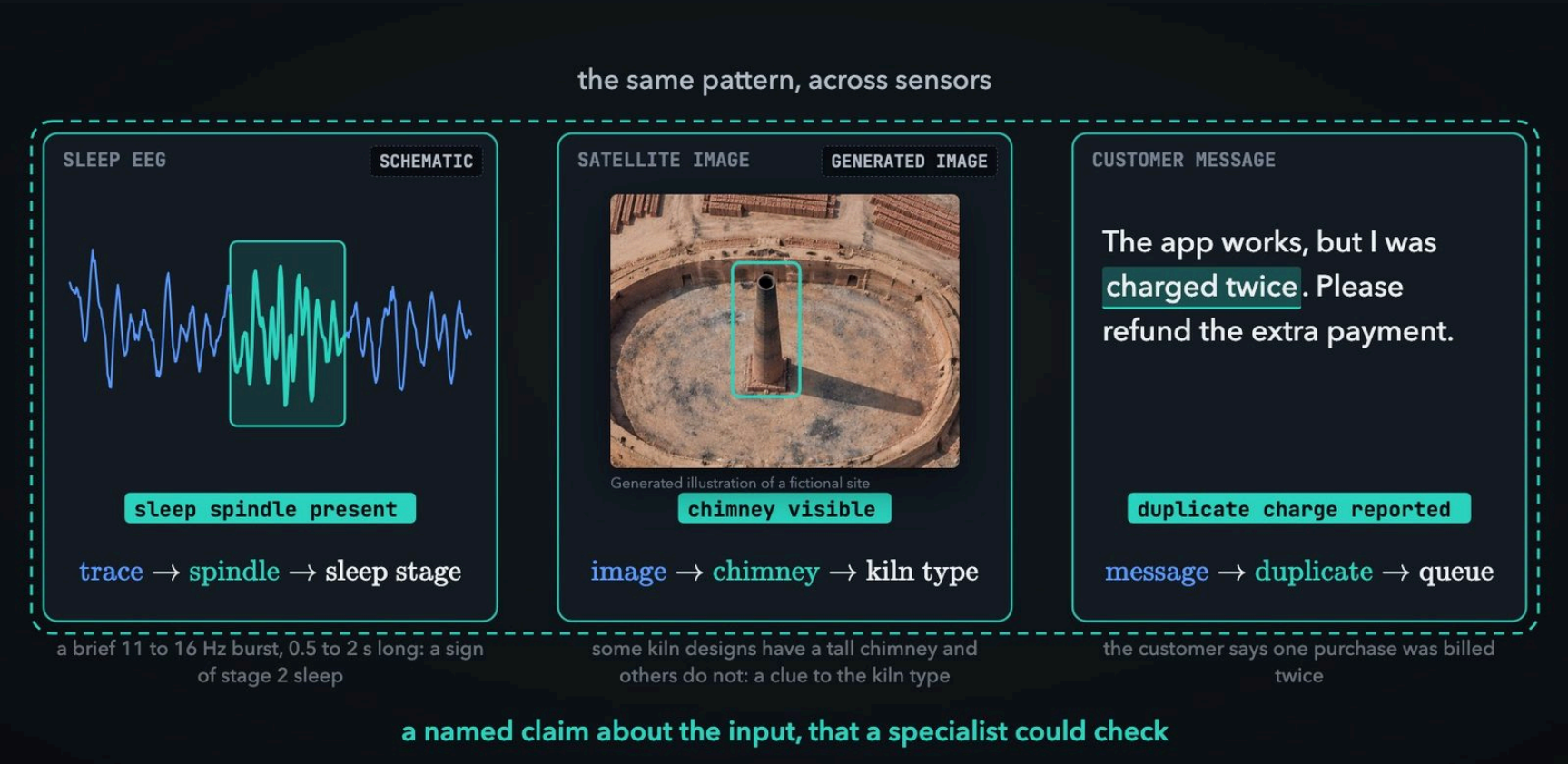
The marked structure in this image

REPRESENTATION

0/1, a score, or a distribution

“Visible chimney” needs a definition, image evidence and a declared encoding.

Meaning, grounding, representation, across sensors



A concept is a definition, a place in the input, and a number with a stated meaning.

Naming concepts and teaching their detection

Pipeline	Who names the concepts?	What teaches detection?
Supervised CBM	Domain definitions	Per-example concept labels
LaBo	LLM proposals + selection	CLIP image-text scores
Label-Free CBM	Filtered concept vocabulary	VLM targets for a learned projection

Task labels still train these classifiers. Discovery can suggest a vocabulary; it does not, by itself, put concepts on the prediction path.

A language model can propose a name while a different model supplies its grounding.

Compare the concept source and the prediction path

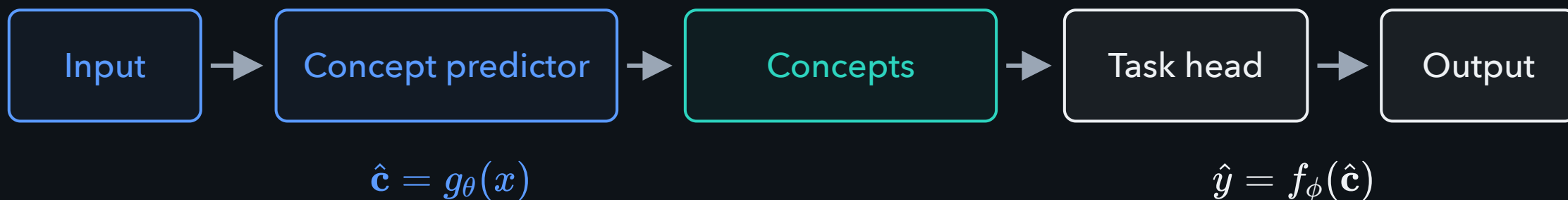
Method	Concept source	Representation	What the head receives
CBM	Human labels	Scalar values	Concepts only
CEM	Concept labels	Embeddings + activity	Concept embeddings
PCBM	External bank	Projection scores	Concepts; residual in PCBM-h
LF-CBM	VLM alignment	Learned activations	Projected concept values
LaBo	LLM vocabulary + CLIP	Similarity scores	Concept scores

TCAV and ACE analyze a predictor. They do not impose this concept-only path.

Check what reaches the head before comparing accuracy or editability.

Training the parts, one strategy at a time

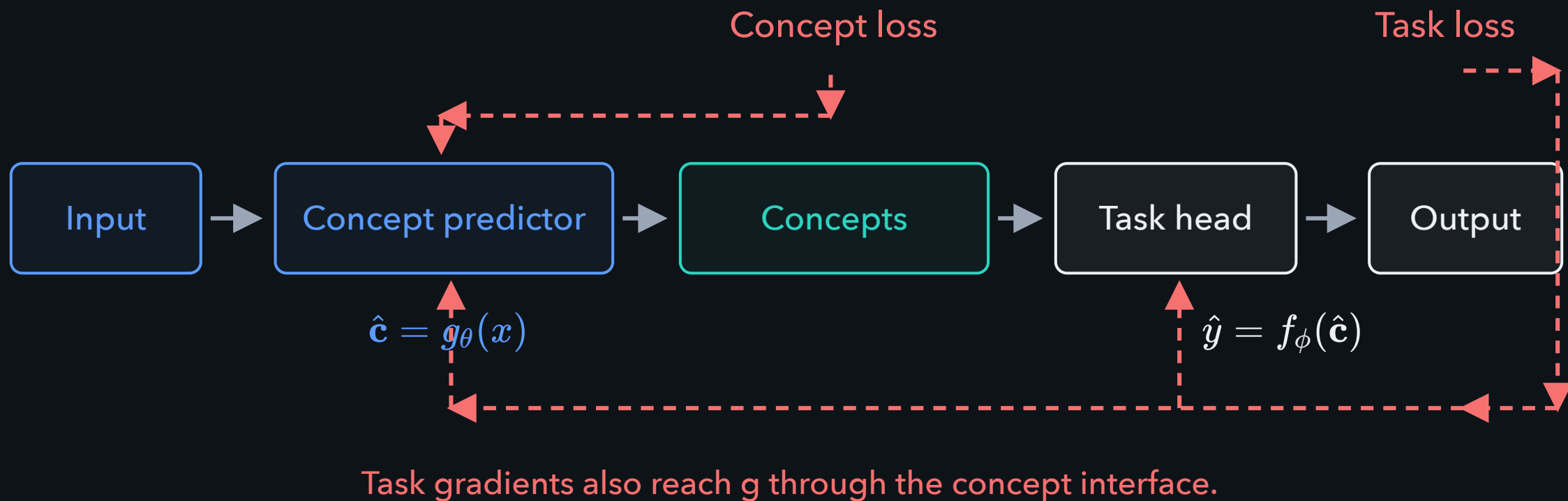
$$\mathcal{L} = \lambda_c \mathcal{L}_c + \lambda_y \mathcal{L}_y$$



Concept targets supervise g . Task targets train both g and f .

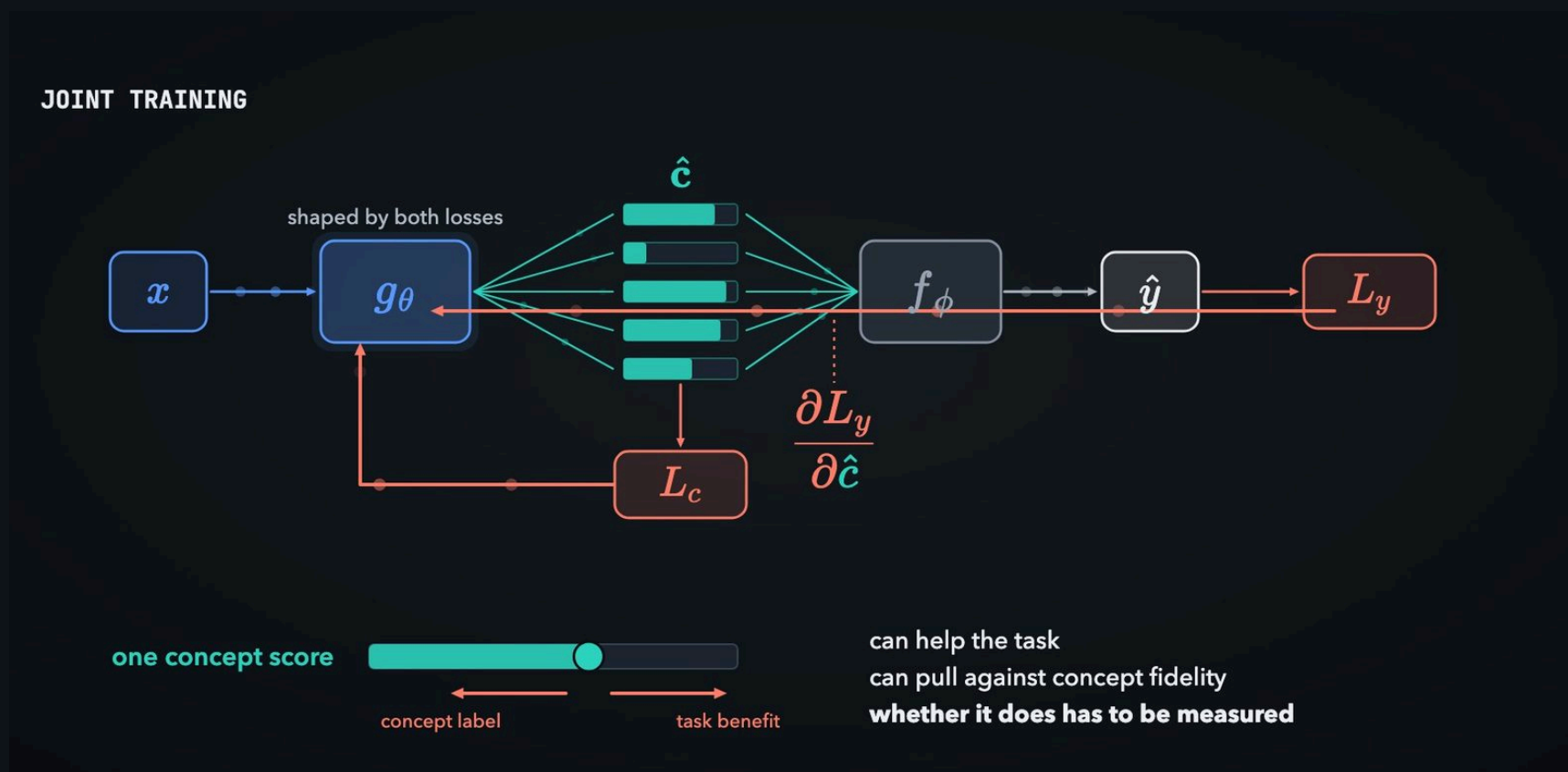
The strategies differ in what the task head sees during training.

Gradient paths in joint training



Concept loss updates $\mathbf{g}$. Task loss updates $\mathbf{f}$ and reaches $\mathbf{g}$ through the concept interface.

Losses, three strategies and joint gradients



Joint training lets the task shape the concepts; whether that hurts fidelity has to be measured.

A linear head: compute one class score at a time

Multiply one row at a time

zebra

$$3 \times 0.1 + 1 \times 0.9 + 0 \times 0.1 = 1.2$$

horse

$$(-1) \times 0.1 + 2 \times 0.9 + (-1) \times 0.1 = 1.6$$

giraffe

$$(-1) \times 0.1 + 0 \times 0.9 + 3 \times 0.1 = 0.2$$

Largest logit: horse

Each row of W gives one class logit. The largest logit selects the class.

What happens when we correct “stripes”?

Same W · only stripes changes

zebra

$$3 \times 1.0 + 1 \times 0.9 + 0 \times 0.1 = 3.9$$

horse

$$(-1) \times 1.0 + 2 \times 0.9 + (-1) \times 0.1 = 0.7$$

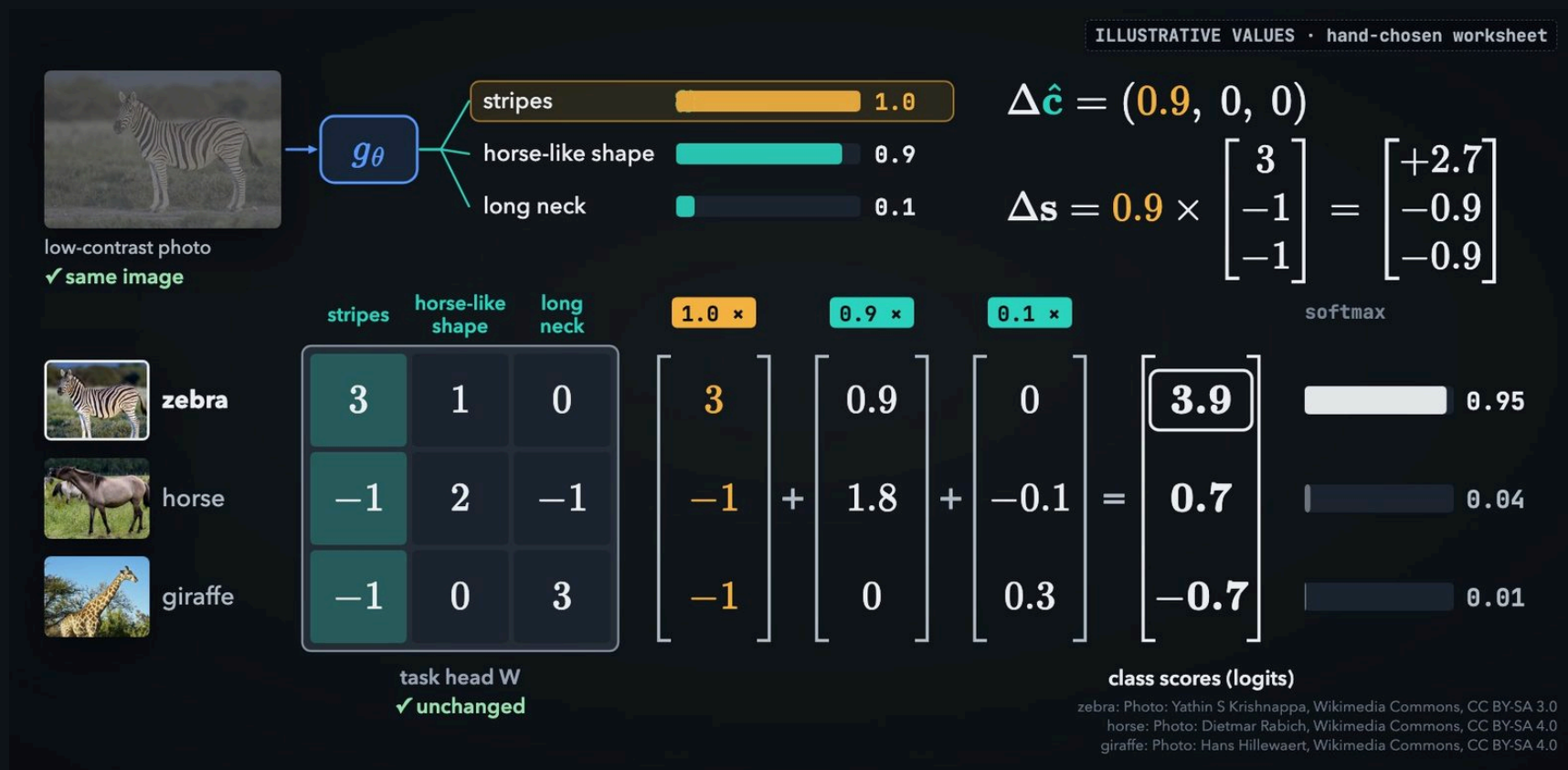
giraffe

$$(-1) \times 1.0 + 0 \times 0.9 + 3 \times 0.1 = -0.7$$

Largest logit: zebra

The weights stay fixed. Only one input to the head changes.

Column view: an edit adds one scaled column



Same image, same W : only one named coordinate moved, so the scores moved by 0.9 times its column.

Try the same three-concept head

Fixed task head

$$W = \begin{bmatrix} 3 & 1 & 0 \\ -1 & 2 & -1 \\ -1 & 0 & 3 \end{bmatrix}$$

Rows: zebra, horse, giraffe
Columns: stripes, horse shape, long neck
Bias = 0

Concept values

stripes

horse shape

long neck

0.10

0.90

0.10

Output

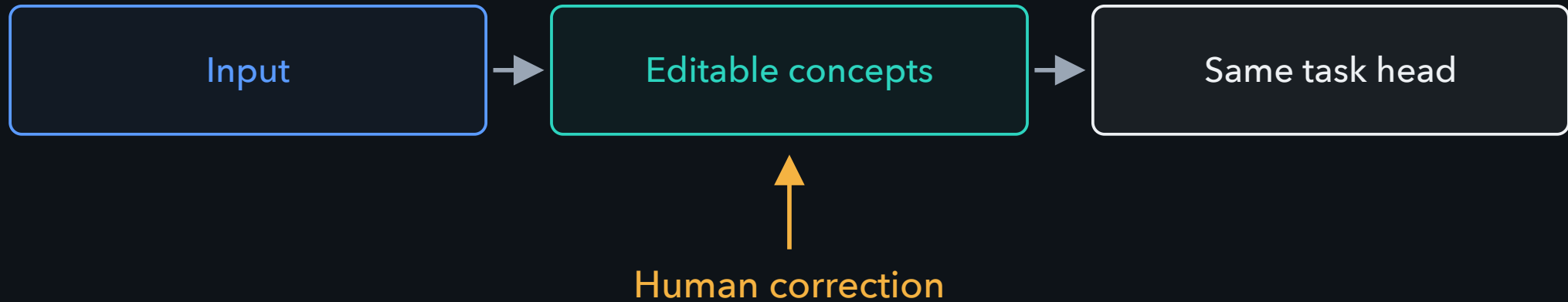
horse

Class	Logit	Softmax
zebra	1.20	35.0%
horse	1.60	52.2%
giraffe	0.20	12.9%

Normalized shares, not calibrated probabilities.

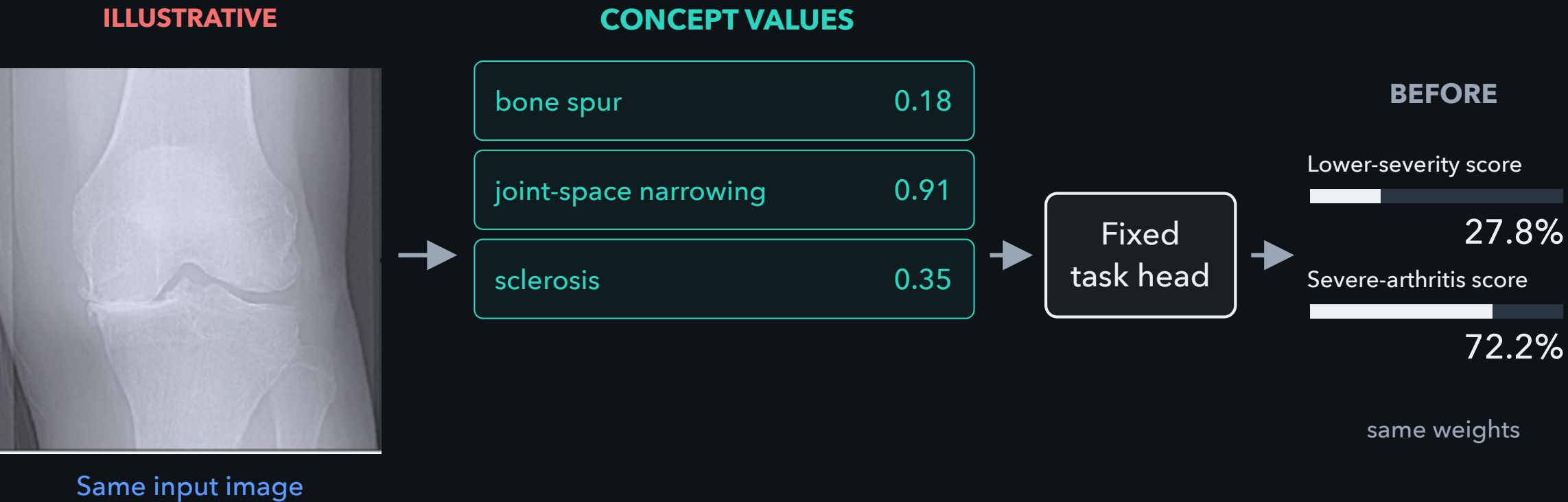
Change one concept and watch both the logits and the softmax shares.

Interventions



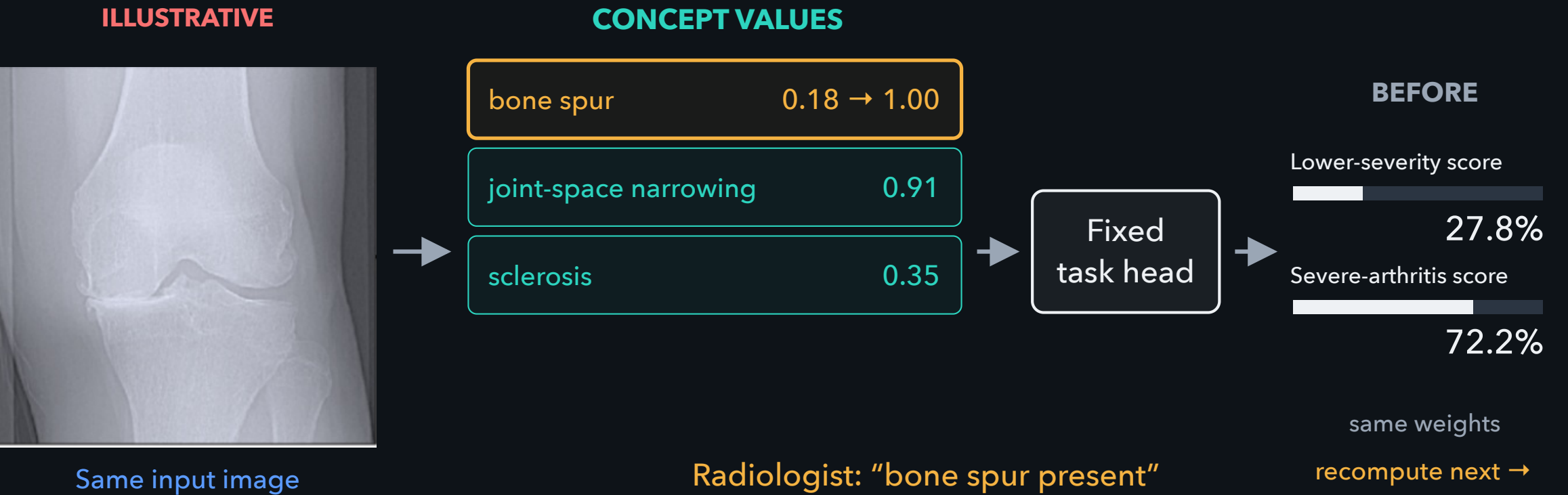
What changes when a person corrects one belief?

X-ray intervention: before the correction



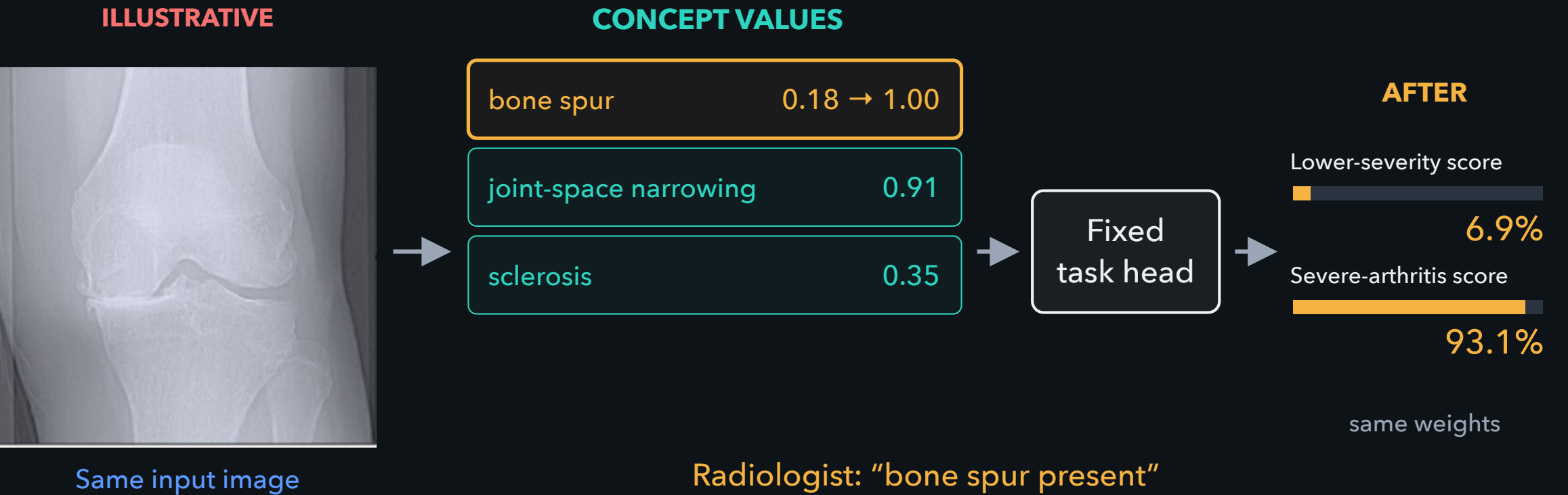
The head uses all three beliefs to compute the severity score.

The radiologist confirms a bone spur



The concept value changes. The old output remains until we recompute.

The same head computes the corrected output



The illustrative severe-class share rises from 72.2% to 93.1%.

Bird intervention: check wingtip colour

ILLUSTRATIVE



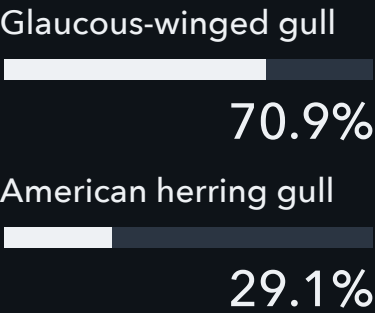
Same input image

CONCEPT VALUES

wingtip: grey	0.80
wingtip: black	0.20
bill: pointed	0.90
head: white	0.85
tail: rounded	0.60

Fixed task head

BEFORE



same weights

Grey and black are two values of one categorical attribute.

Correcting wingtip colour changes the ranking

ILLUSTRATIVE



Same input image

CONCEPT VALUES

wingtip: grey 0.80 → 0.00

wingtip: black 0.20 → 1.00

bill: pointed 0.90

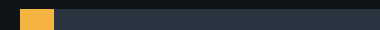
head: white 0.85

tail: rounded 0.60

Fixed
task head

AFTER

Glaucous-winged gull



9.0%

American herring gull



91.0%

same weights

Observer: "wingtip colour is black"

The group changes from grey/black = 0.8/0.2 to 0/1. Other attributes stay fixed.

Kiln intervention: check the chimney

ILLUSTRATIVE



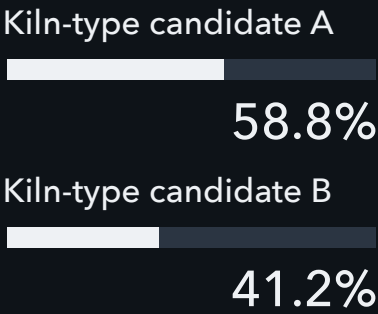
Same input image

CONCEPT VALUES

chimney visible	0.21
elongated footprint	0.88
circular footprint	0.12
smoke plume	0.30

Fixed task head

BEFORE



same weights

The analyst checks the visible structure against the chimney concept.

The chimney correction changes the kiln-type score

ILLUSTRATIVE



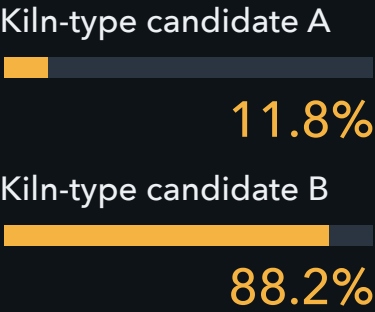
Same input image

CONCEPT VALUES

chimney visible	0.21 → 1.00
elongated footprint	0.88
circular footprint	0.12
smoke plume	0.30

Fixed task head

AFTER

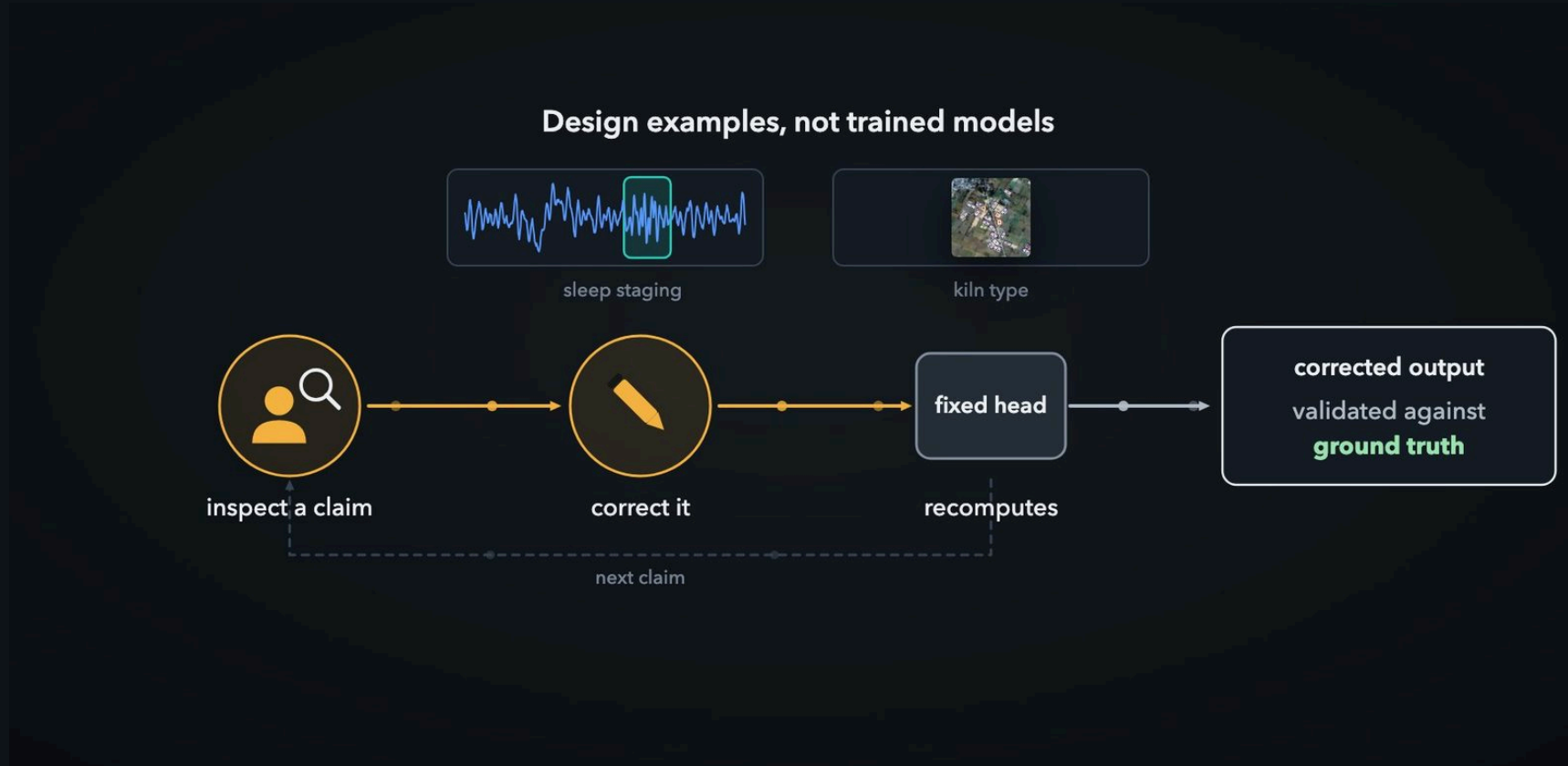


same weights

Analyst: "chimney is visible"

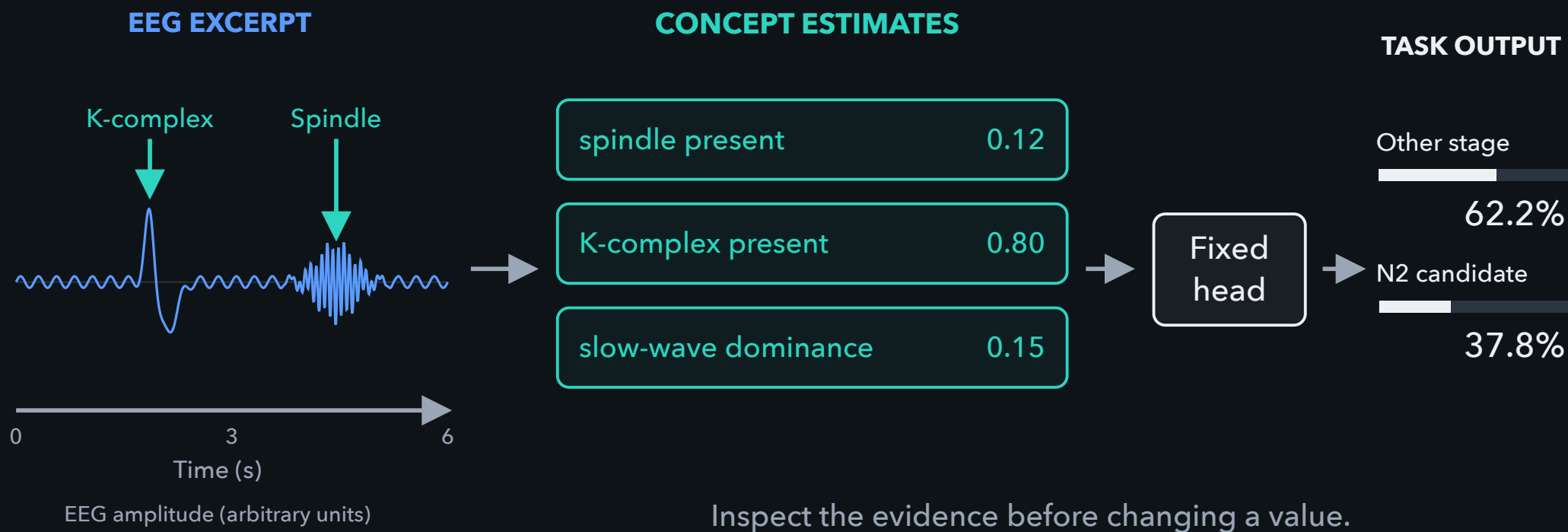
Only chimney visibility changes from 0.21 to 1.00.

Observability depends on the sensor



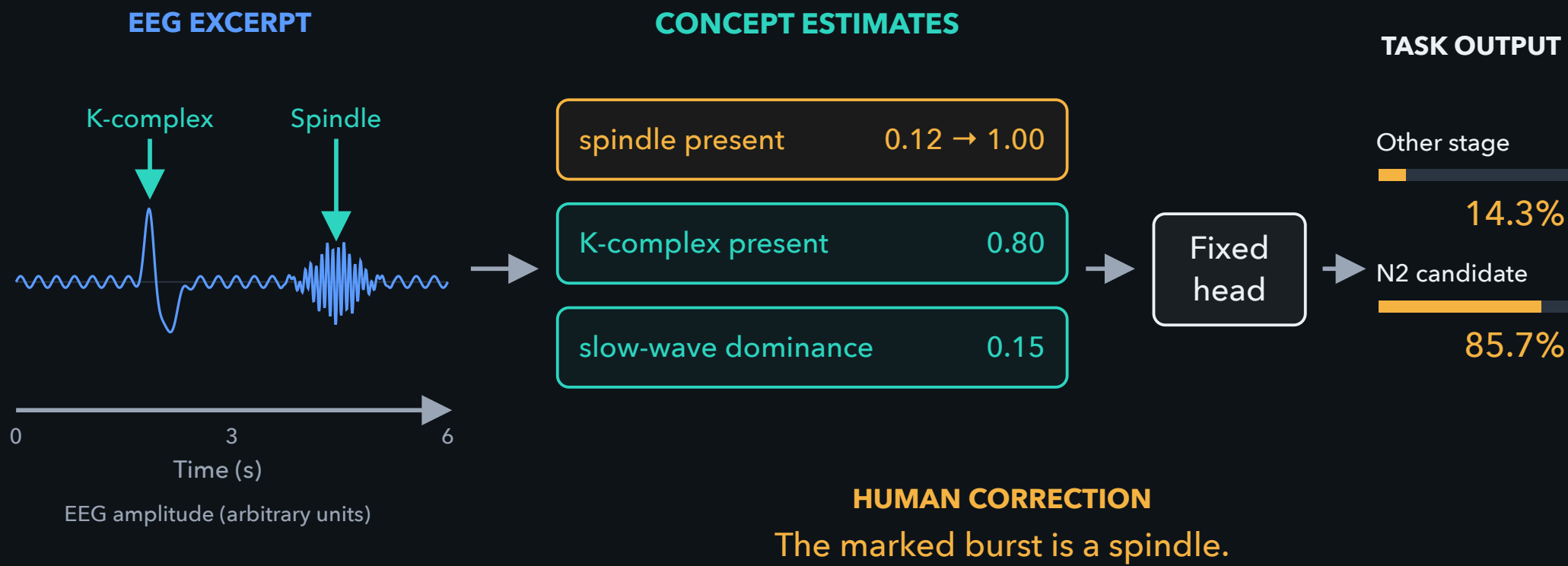
At 10 m per pixel, record "chimney visible" as unknown, not absent.

A sleep expert checks the spindle



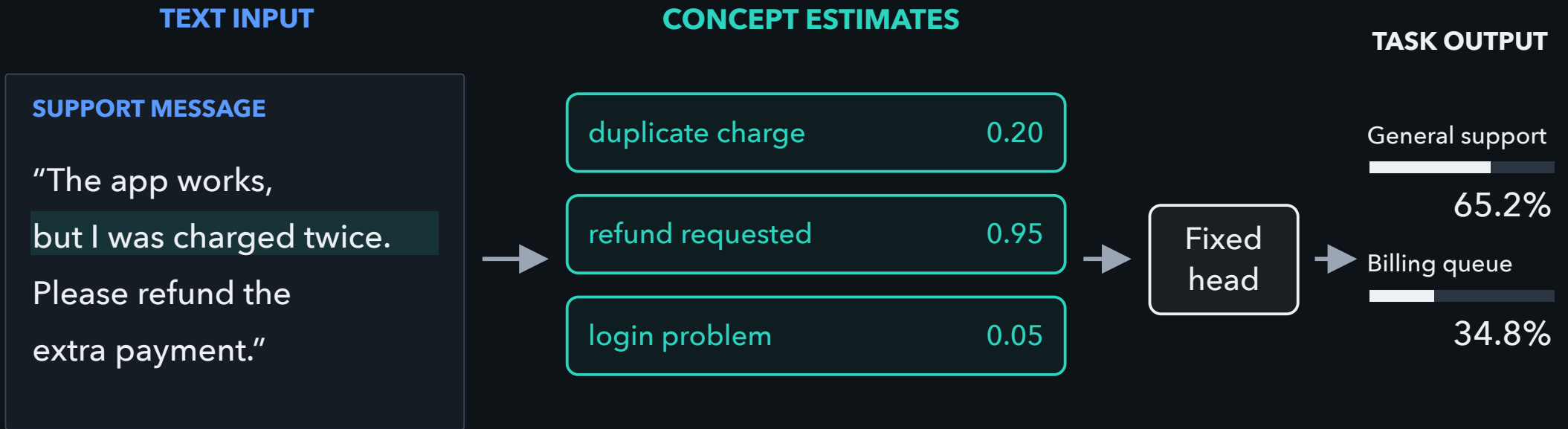
Compare the marked input evidence with the concept estimate.

The corrected spindle changes the stage score



The input and task-head weights stay fixed. Amber marks the corrected concept.

A reviewer checks the duplicate-charge claim

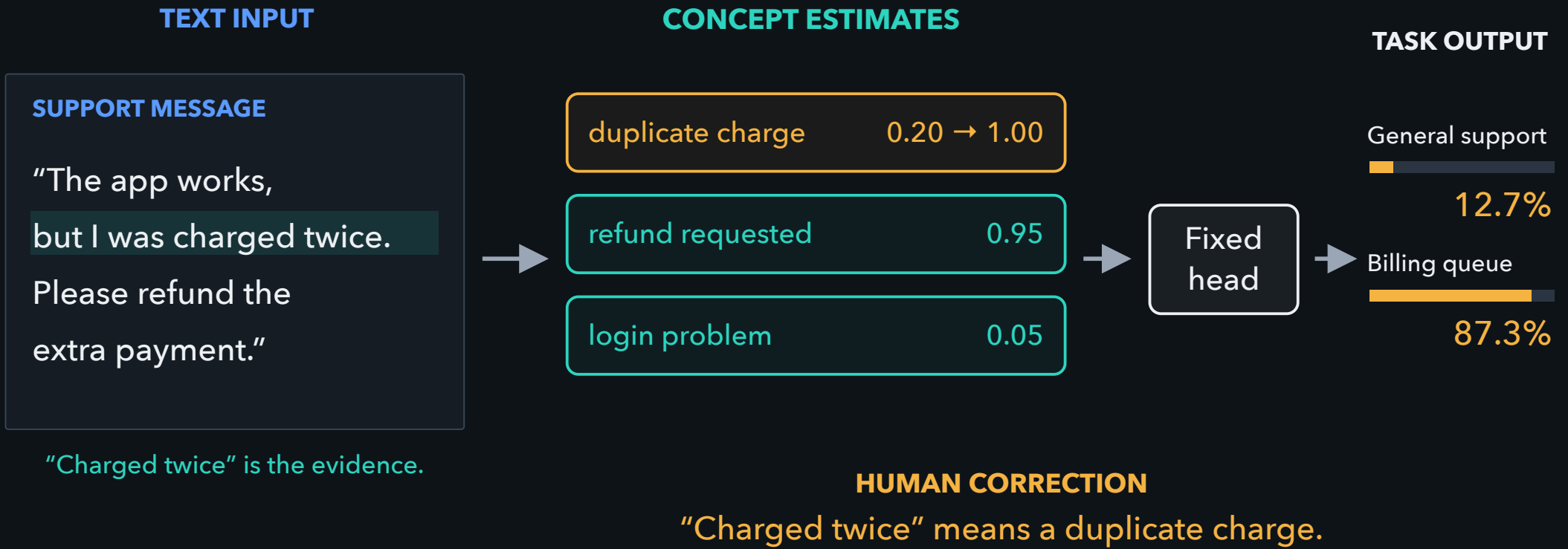


"Charged twice" is the evidence.

Inspect the evidence before changing a value.

Compare the marked input evidence with the concept estimate.

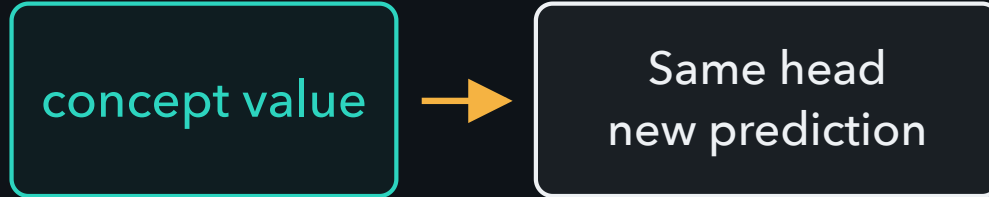
The same text now reaches the billing queue



The input and task-head weights stay fixed. Amber marks the corrected concept.

Local corrections and global model edits

LOCAL: ONE EXAMPLE



$$\hat{c}_j \leftarrow c_j^{(\text{human})}$$

Correct "spindle present" for this window.

GLOBAL: THE MODEL RULE



$$w_j \leftarrow 0$$

Remove reliance on a spurious concept.

A local correction changes an input to the head. A global edit changes the head's weights.

Choosing which concept to correct

Oracle

Know the true concepts
↓
Rank absolute prediction errors
↓
Correct largest error first

Random

No concept ranking
↓
Sample an order
↓
Correct the chosen values

Human

Inspect available evidence
↓
Choose a useful, reliable edit
↓
Pay attention and annotation cost

Selection policy and feedback quality both affect the result.

Will a correct concept edit improve the prediction?

Same message: "I was charged twice. Please refund the extra payment."

Required route: Billing queue

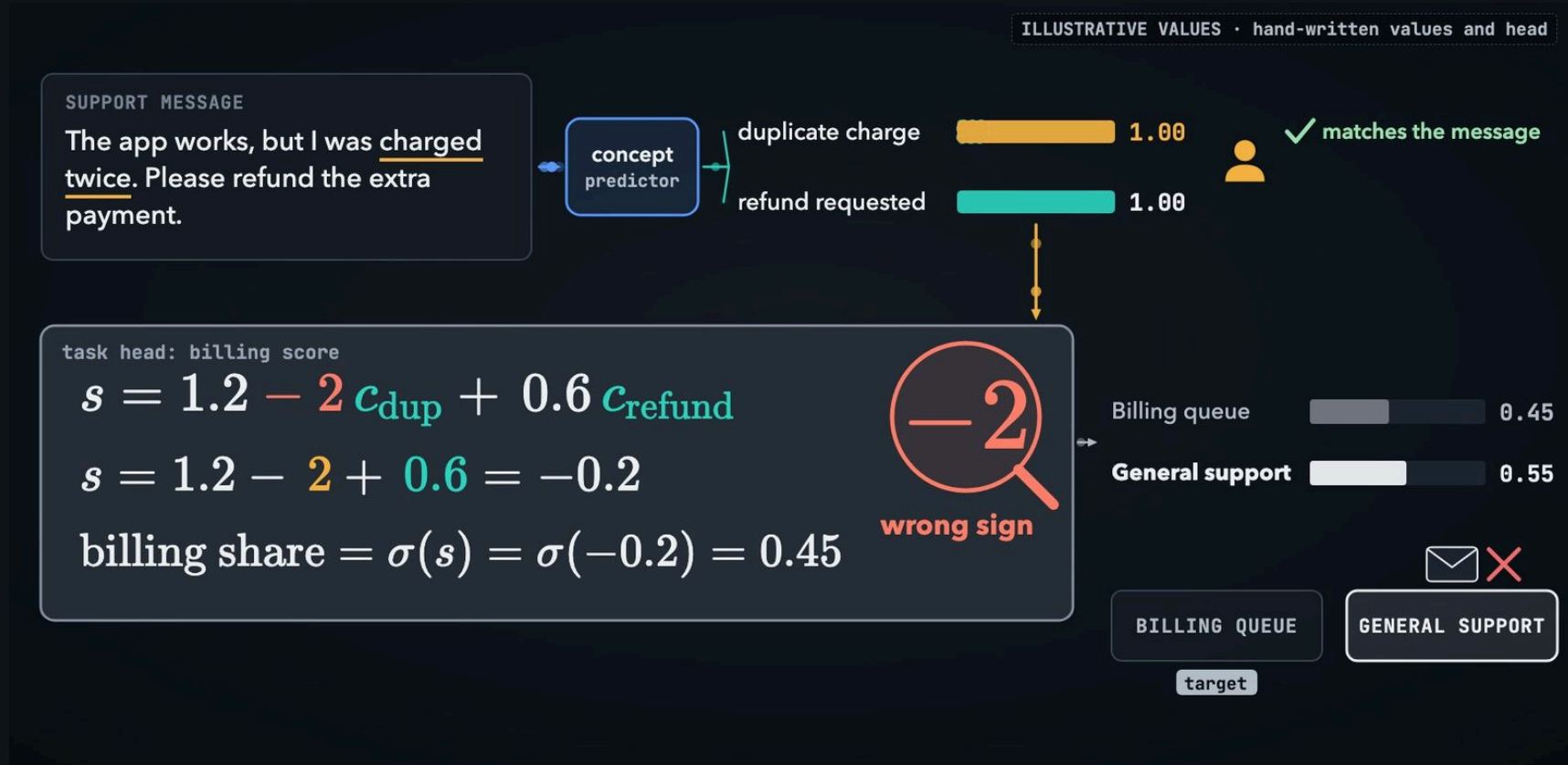
$$z_{\text{billing}} = 1.2 - 2\hat{c}_{\text{duplicate}} + 0.6\hat{c}_{\text{refund}}, \quad z_{\text{general}} = 0$$



The concept is now correct. The faulty head reverses the decision.

Predict the effect before recomputing the fixed head.

Worked: a correct edit through a faulty head



Correcting a concept does not repair a wrong decision rule.

Failure modes

Wrong estimate

large: 0.78; truth: 0

Missing concept

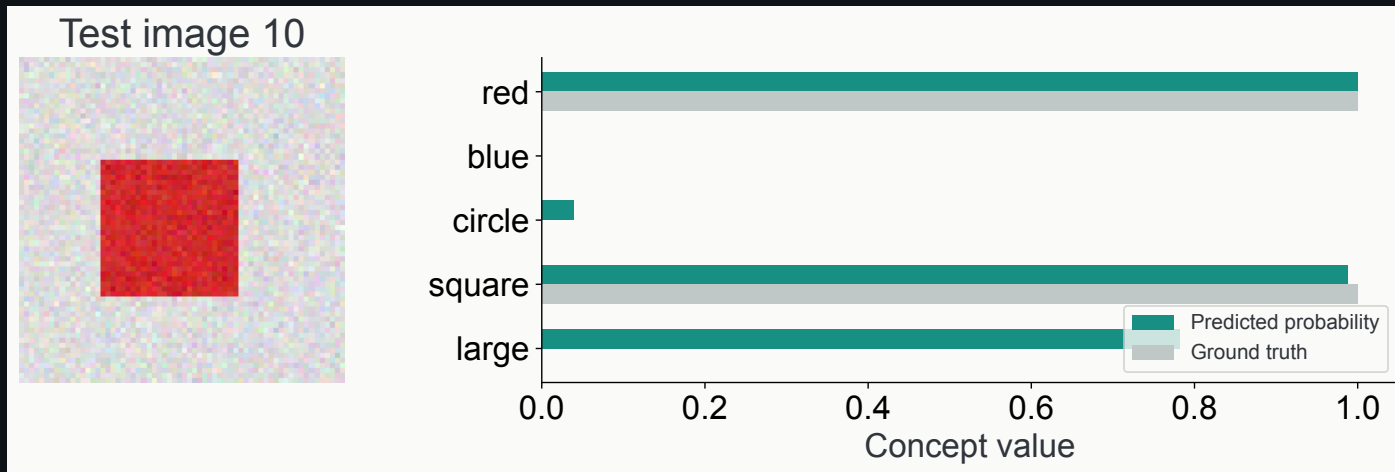
shape was never recorded

Extra information

one score also reveals class

Which failure can a concept correction repair?

Concept prediction error



The model says
large.

The held-out object is small.
Its size concept is naturally wrong.

The model's size estimate is wrong on this held-out image.

Does the detector understand the concept?

SAME PHRASE

"I was charged twice.
Please refund the extra payment."



DIFFERENT CLAIM

duplicate charge: PRESENT

Expected concept label

"I was not charged twice.
The alert was a display error."



duplicate charge: ABSENT

Expected concept label

A keyword detector would fail this authored contrast test.

Test a change in meaning while keeping the tempting keyword.

A correct concept can come from the wrong evidence

Original scene



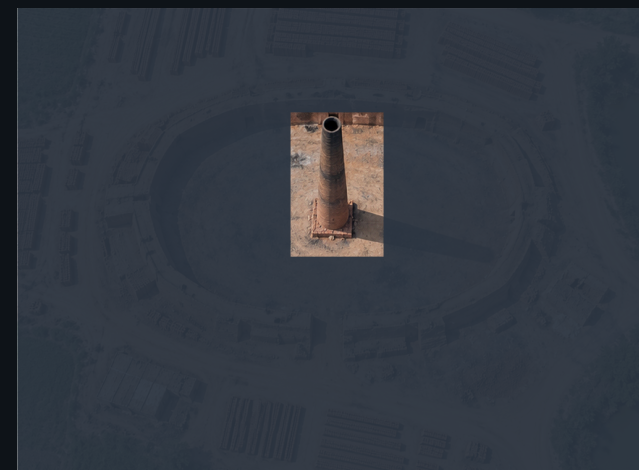
Detector: present (correct)

Mask chimney evidence



Allow "cannot tell"

Mask surrounding context



Does detection persist?

Compare both masks plus matched controls. These are proposed tests, not outputs.

A correct "chimney present" prediction does not show which evidence produced it.

An incomplete vocabulary loses a class distinction



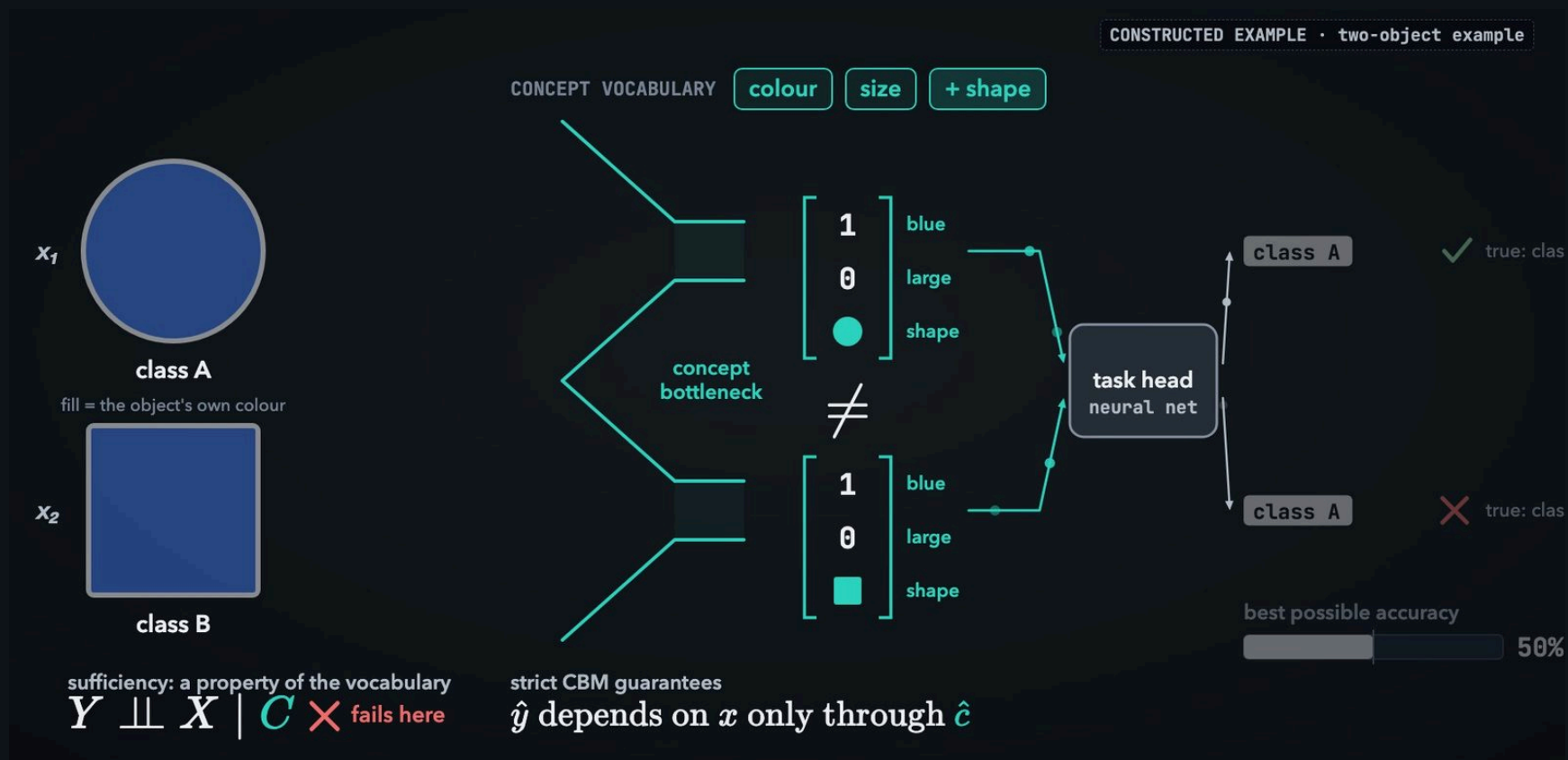
$$c(x_1) = c(x_2), \quad y_1 \neq y_2$$

$$f(c(x_1)) = f(c(x_2))$$

A more powerful deterministic head cannot separate identical inputs.

A task head cannot recover a distinction that the concept vector has removed.

Shape disappears at the bottleneck



No head can recover a distinction that never reached it.

When do concepts contain enough task information?

$$Y \perp X \mid C$$

Once the true concepts are known, the input adds no further information about the true target.

$$\hat{Y} = f(\hat{C})$$

Architectural routing

A different statement about a model.

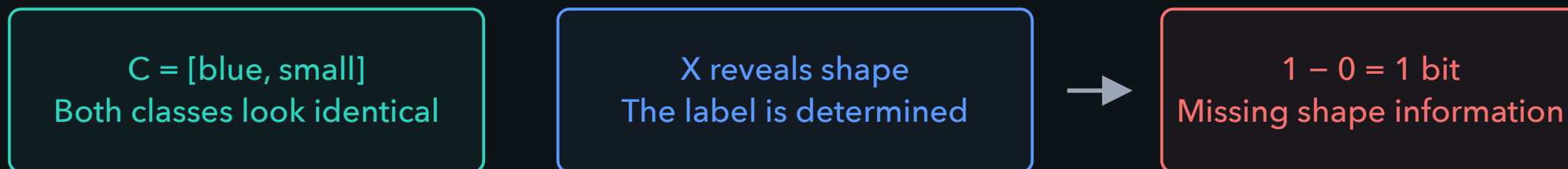
Sufficiency depends on the task and data distribution.

True-target sufficiency is a property of the representation and distribution.

Missing task information can be measured in bits

$$I(Y; X \mid C) = H(Y \mid C) - H(Y \mid X, C)$$

Authored population: equal numbers of blue circles and blue squares



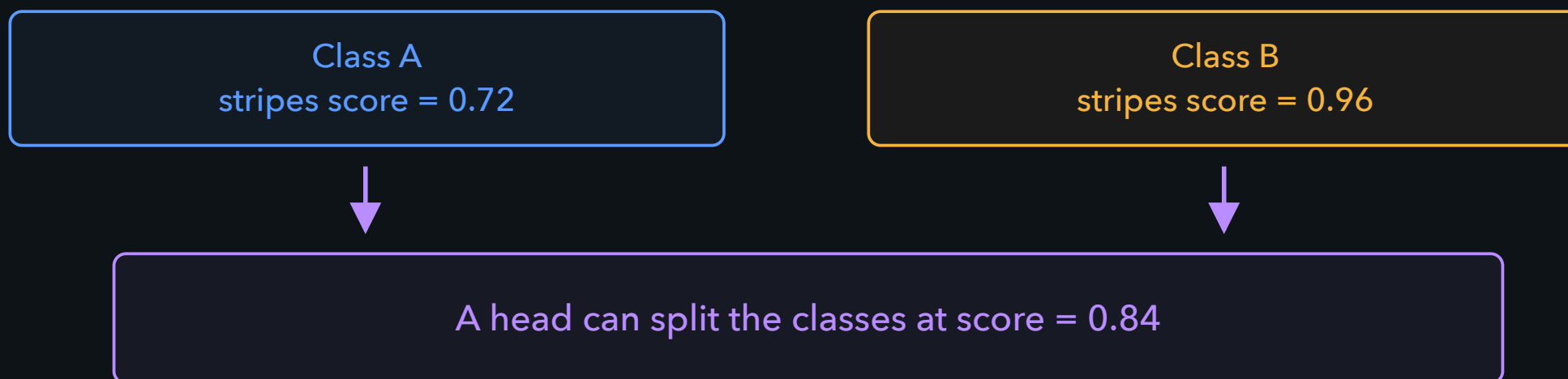
$$I(Y; X \mid C) = 0 \iff Y \perp X \mid C$$

Discrete variables, finite entropies; not an estimate from the trained model.

Positive conditional mutual information means X still informs Y after C is known.

Leakage through a soft concept score

TRUE SEMANTICS: stripes present in both classes

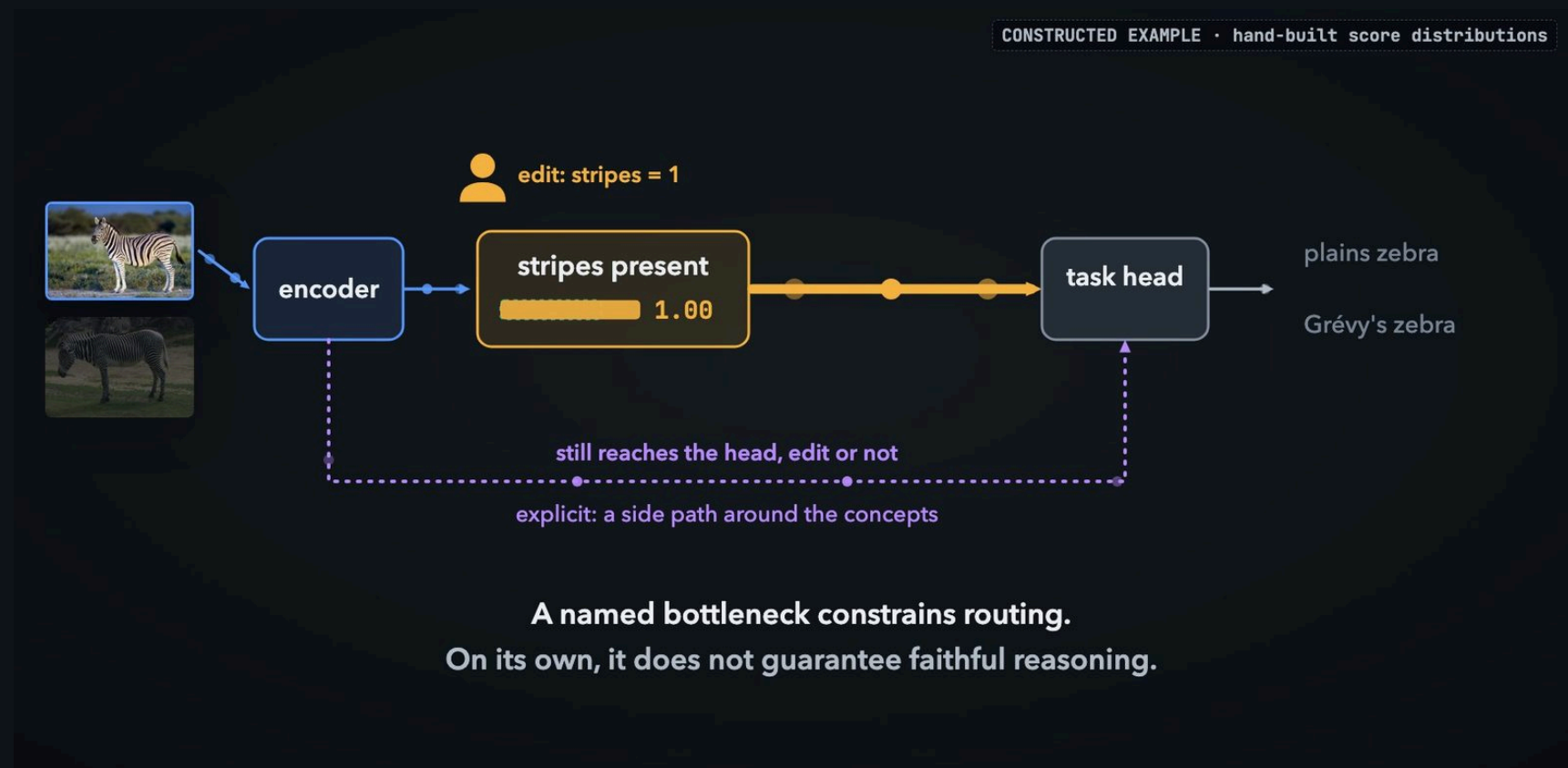


Hidden channel: exact decimals inside a named score.

An explicit residual bypass is a separate route around the interface.

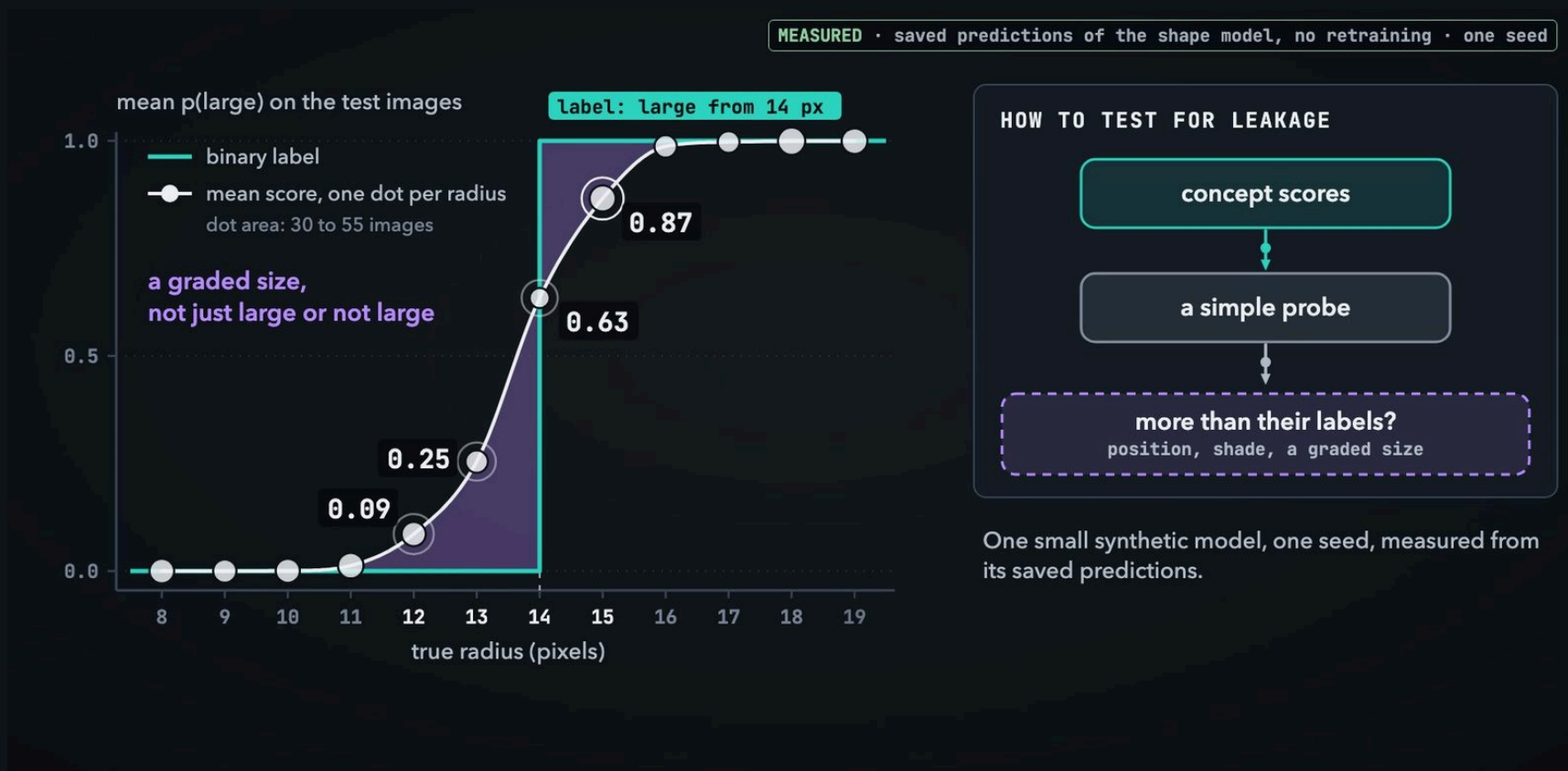
A score can carry extra information even with no connection around the bottleneck.

Leakage, step by step



A named bottleneck constrains routing; it does not guarantee faithful reasoning.

Does a trained model leak?



The measured size score carries a graded size, but not unrelated nuisances.

Reducing leakage

REDUCING LEAKAGE

1 Hard concepts

Pass only 0 or 1 to the head, so nothing extra can travel.



cost: the head loses graded evidence.

2 Independent training

Fit the head on true binary concepts.

It never learns to read hidden detail in the scores.



3 Remove or report side paths

Remove side paths around the concepts,
or report them clearly when added for accuracy.



4 Measure

Probe what the scores carry beyond their labels.

Test whether interventions actually help, on held-out data.



See Havasi et al. 2022, on addressing leakage in concept bottleneck models.

Hard concepts, independent training, no hidden side paths, and measurement.

The human can also supply the wrong correction

Authored case: the true class is HORSE; stripes are absent.



Same W . Same calculation. This time the feedback is wrong.

Allow uncertainty, abstention and reversal of a correction.

Evaluate feedback quality and allow a reviewer to say "cannot tell."

A model intervention changes a model input

Model intervention



Change the bone-spur input

How does this model's output change?

Real-world causal intervention

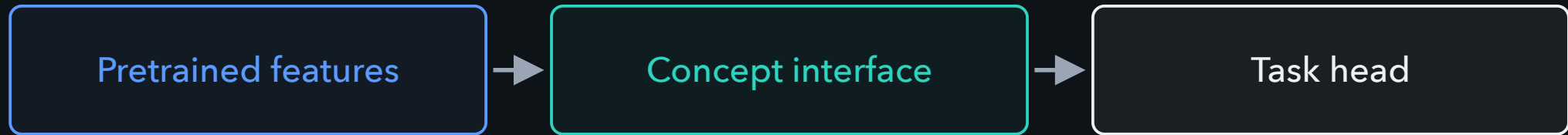


Change a physical mechanism

Would a clinical outcome change?

Changing a bone-spur variable does not establish a causal effect on disease.

Newer methods



Change the concept source or its representation.

How can we change the source or representation of concepts?

What newer methods change

	supervised CBM	CEM	PCBM	label-free	LaBo
concept source	people label every concept				
detector	a trained concept predictor				
representation	one number per concept				
what reaches the head	only the concept values				

24 train

zebra

horse

giraffe



36 Wikimedia Commons photographs (CC BY-SA / CC BY / public domain)

12 test

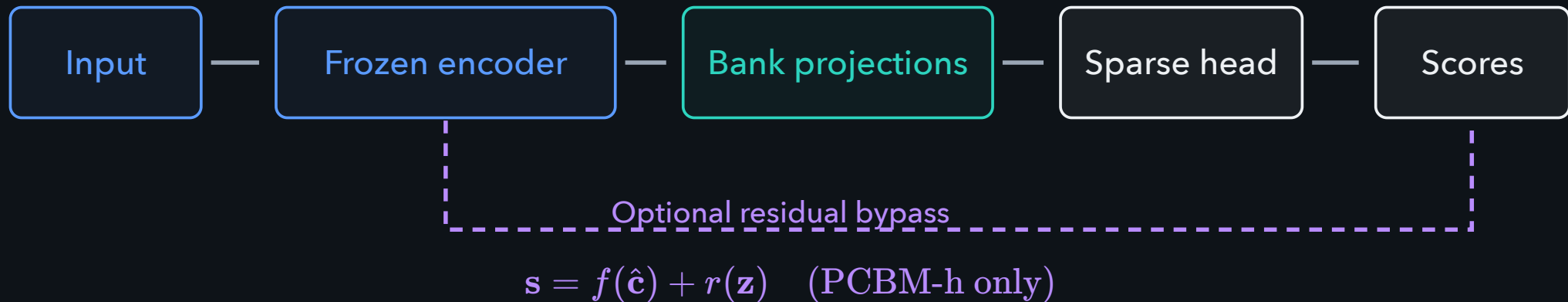


the classes from Part 4

Newer variants change the concept source, the detector, the representation and what reaches the head.

PCBM builds a concept model on pretrained features

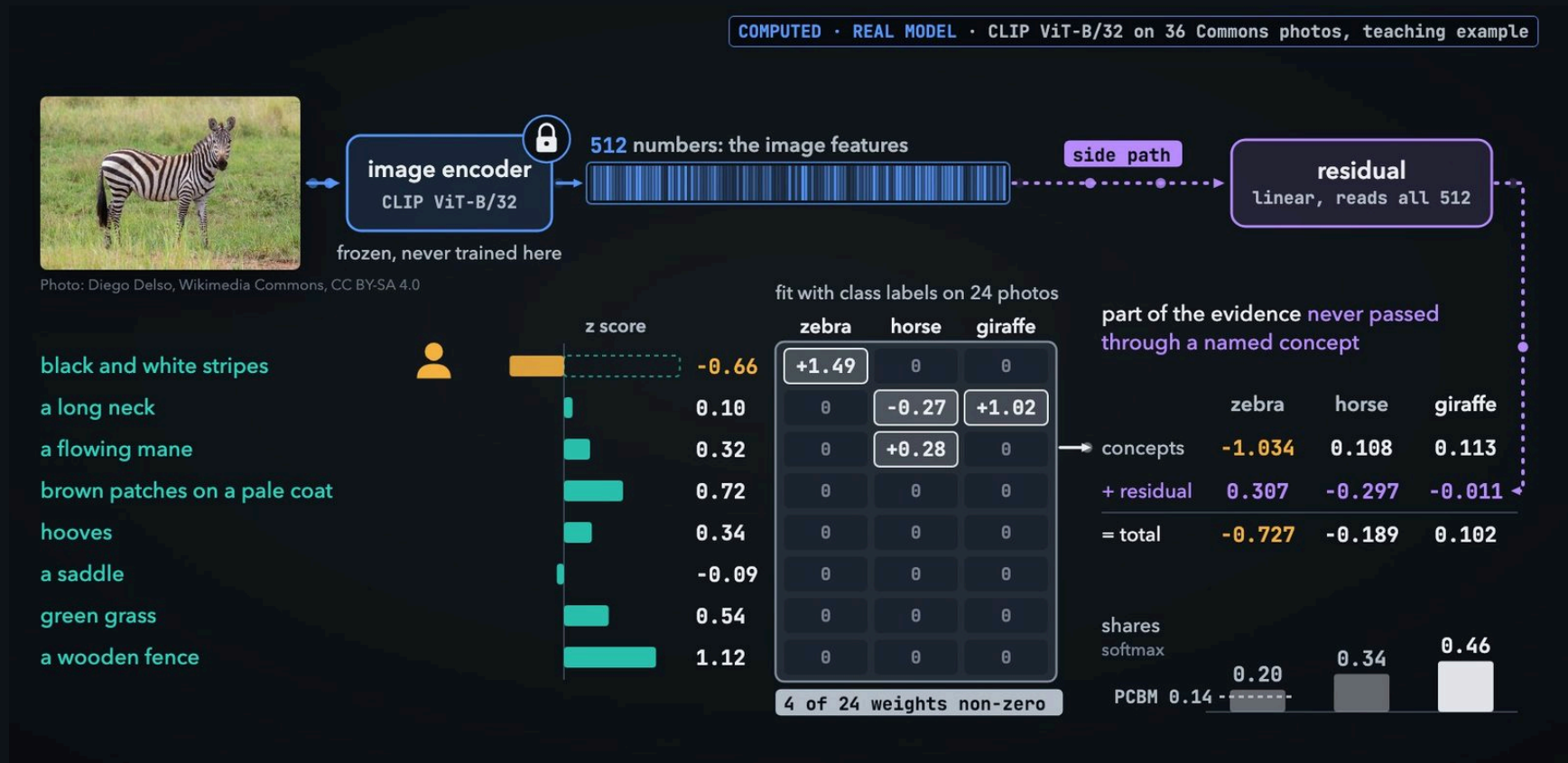
External concept bank defines projection directions



Base PCBM: concept scores only. PCBM-h: also a feature residual.

The base model uses concepts. PCBM-h also has a feature residual.

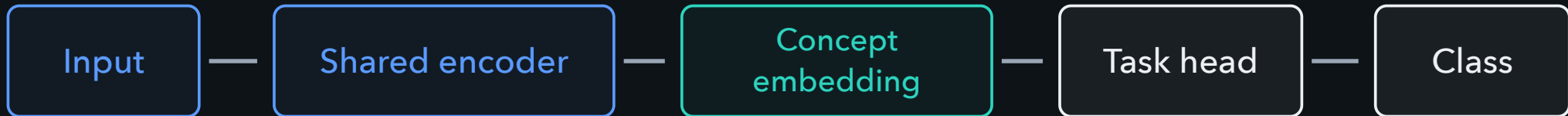
Worked: post-hoc CBM on real CLIP features



A residual side path keeps part of the evidence outside the named concepts.

CEM represents each concept with an embedding

Concept labels supervise the activity probability

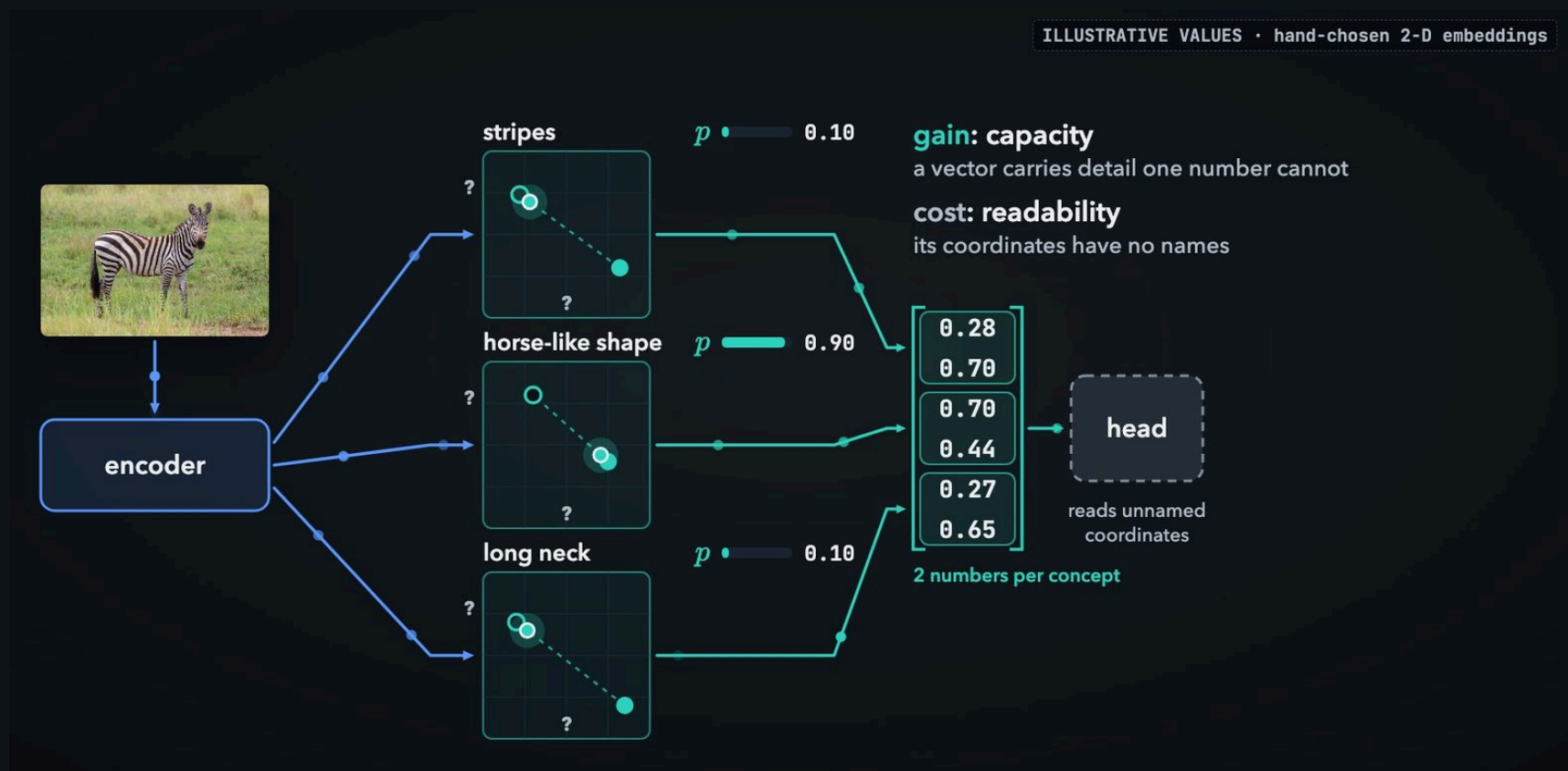


$$\mathbf{e}_j = \hat{p}_j \mathbf{e}_j^+ + (1 - \hat{p}_j) \mathbf{e}_j^-$$

Edit the mixture probability; embeddings retain extra capacity.

A person edits the mixing probability between active and inactive concept embeddings.

Worked: a concept embedding model



An intervention sets p ; the vector carries more than one number, but its coordinates are unnamed.

CLIP scores image-concept text pairs



Image encoder

u



Text encoder → normalized text vectors

A cosine similarity needs validation before it can be treated as a concept probability.

LaBo uses language to propose a concept vocabulary

LLM proposes names; CLIP scores image-text agreement



$$s_j = \text{COS}(\mathbf{z}_{\text{image}}, \mathbf{z}_{\text{text},j})$$

Edit concept scores or head weights; similarity is not a probability.

The language model proposes concepts. CLIP measures image-text agreement.

Worked: LaBo, language in a bottle



Naming a concept is not the same as detecting it.

The variants side by side

	supervised CBM	CEM	PCBM / PCBM-h	label-free CBM	LaBo
concept source	people label every concept	people label every concept	concept bank: examples or text	language model, then filters	language model sentences, 3 per class
detector	network trained on concept labels	network trained on concept labels	frozen encoder, projected onto concept directions	ResNet-50 + projection fitted to CLIP targets	CLIP itself
representation	one probability per concept	two embeddings mixed by p	projection score per concept	learned activation per concept	CLIP similarity × 100
what reaches the head	scalars	vectors	scalars plus a residual	learned projections	raw CLIP scores
grounding automatic?	no	no	no	no	no



Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

"black and white stripes"



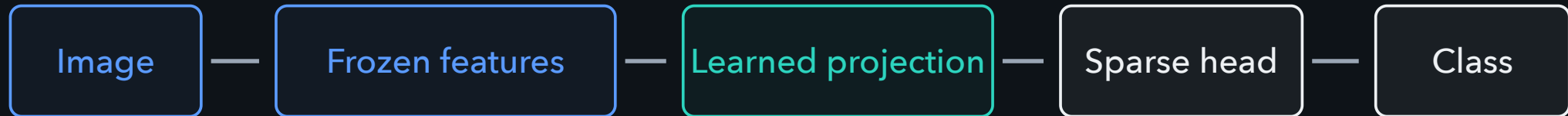
a phrase or a score does not show that the model sees it

naming a concept $\neq$ detecting it: check grounding

Each change makes concepts cheaper or predictions more accurate; none makes grounding automatic.

Label-Free CBM learns a concept projection

VLM targets train the concept projection

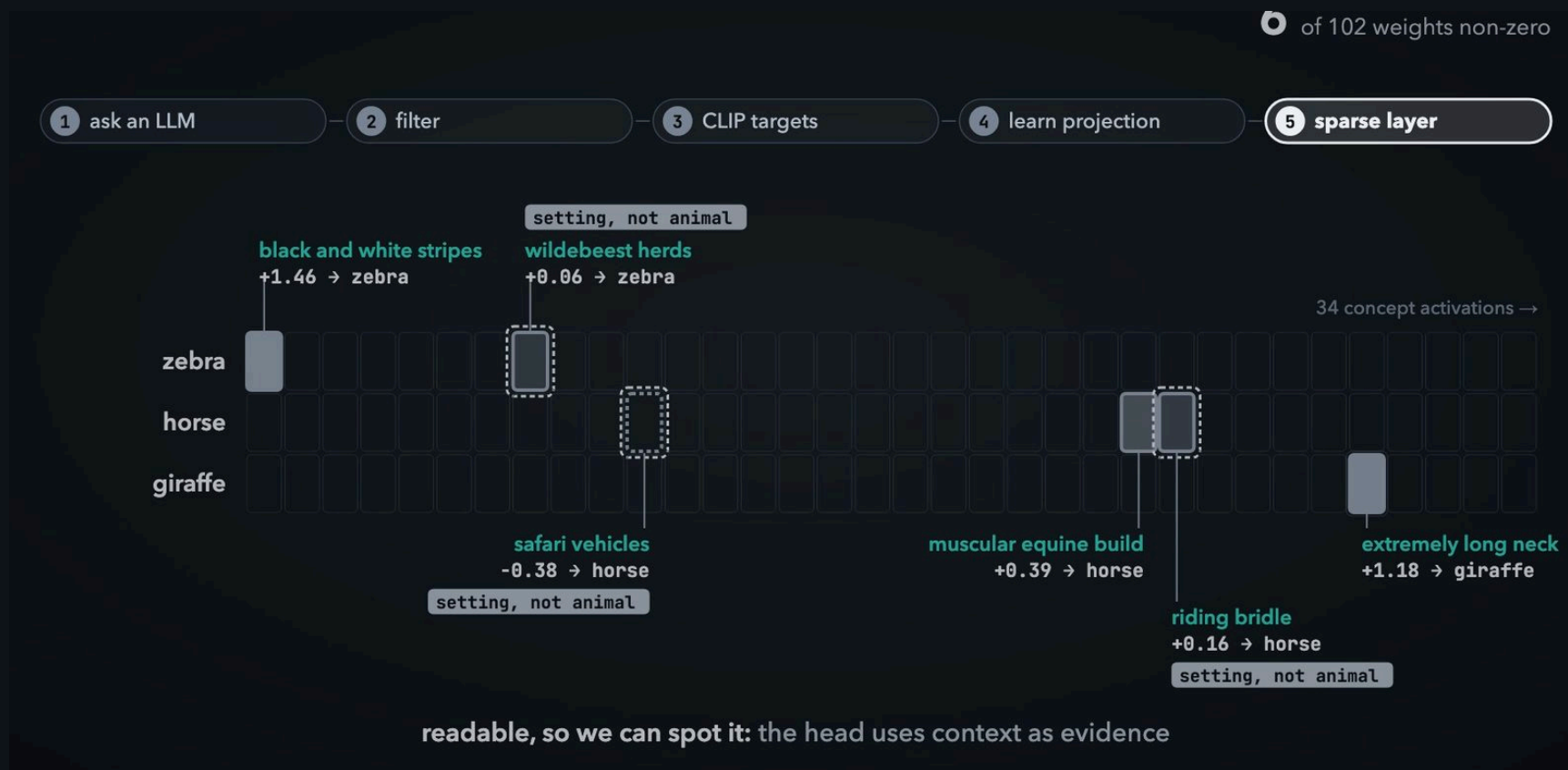


$$\mathbf{z} \mapsto \hat{\mathbf{c}} \mapsto \hat{y}$$

Task labels fit a sparse head. Deployment uses the learned projection.

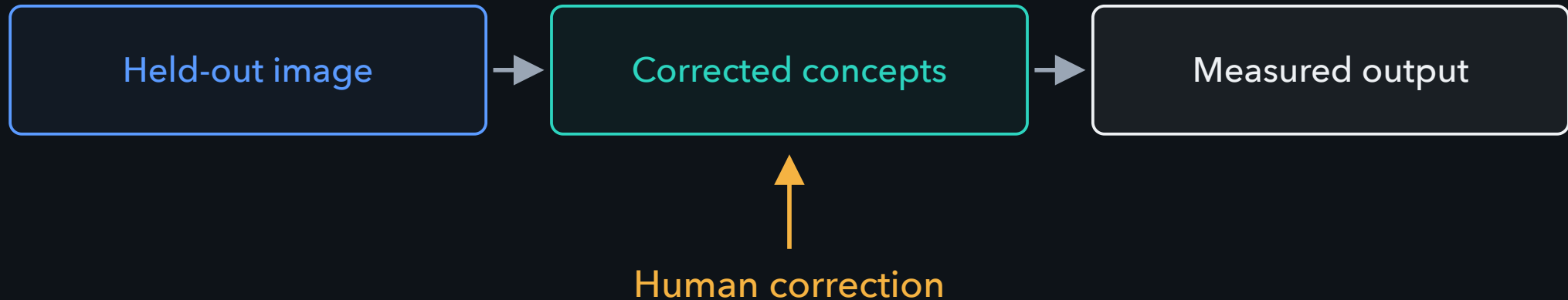
VLM signals train the concept projection. Task labels train the sparse predictor.

Worked: a label-free CBM in five steps



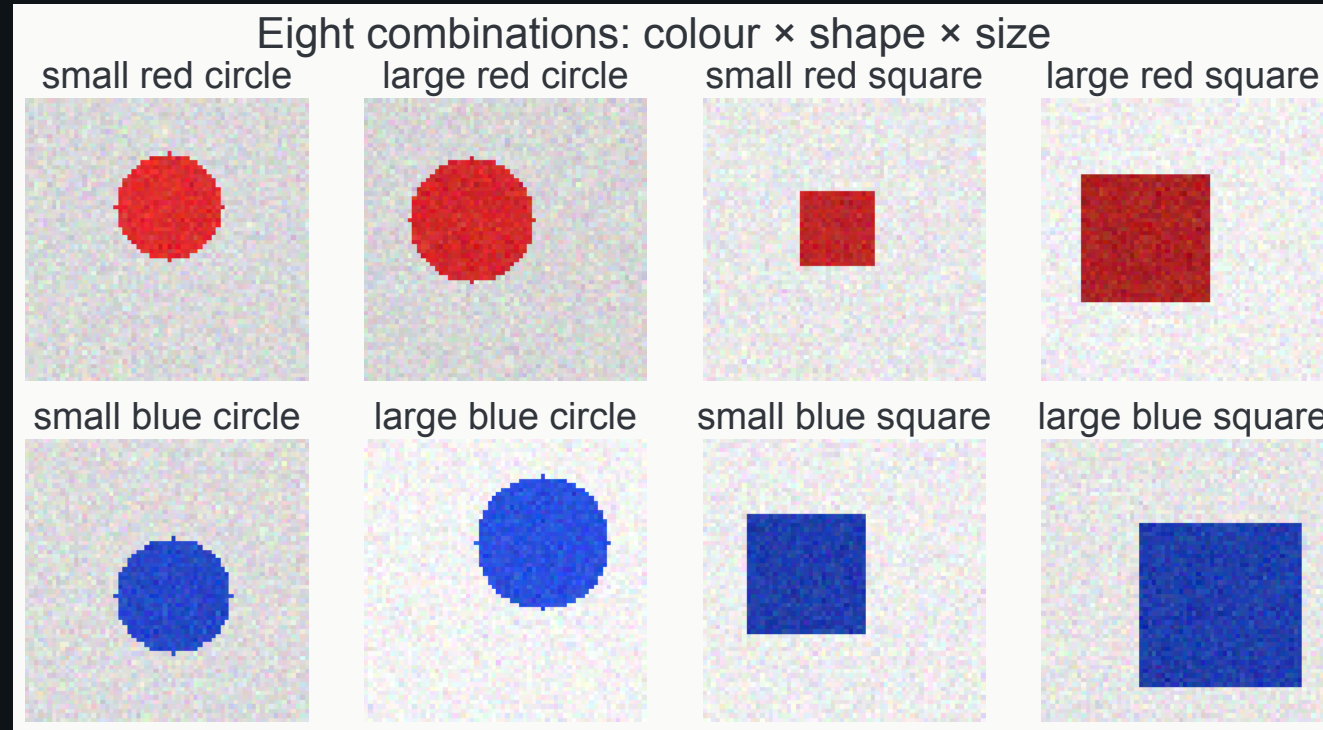
Readable concepts let us see when the head uses the setting, not the animal.

Measured evidence



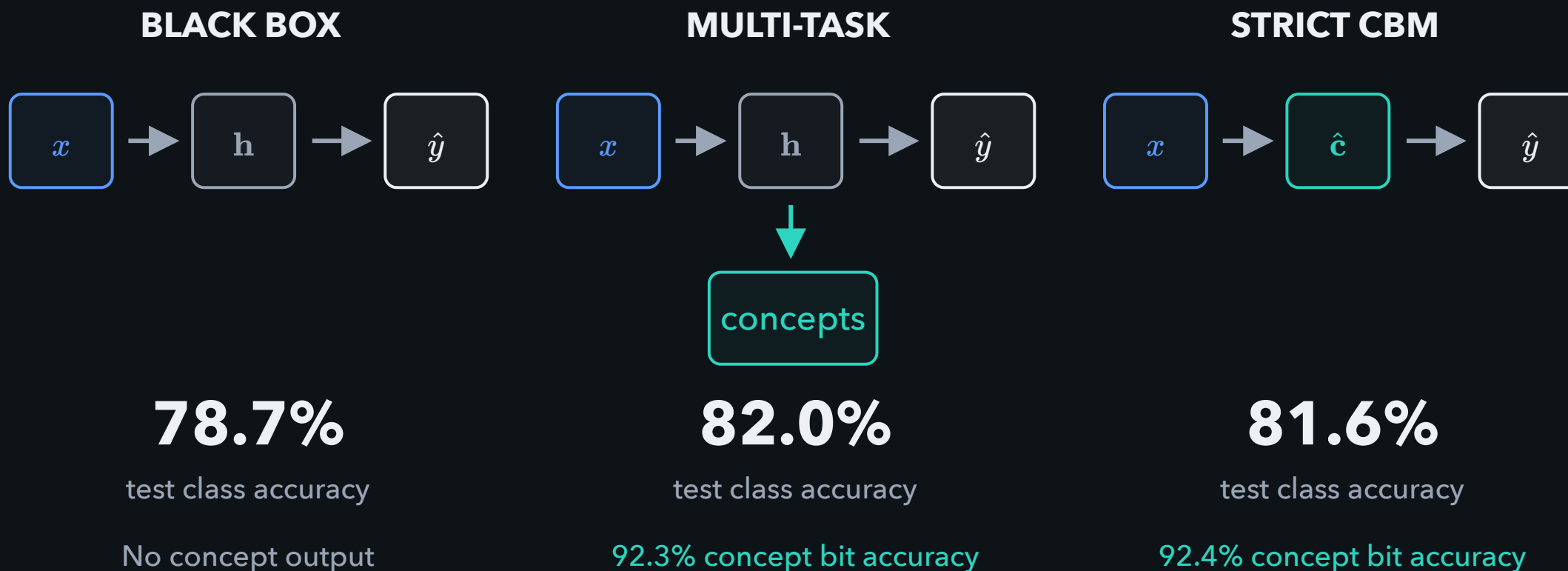
Does the trained model respond to a correction?

The measured shape experiment



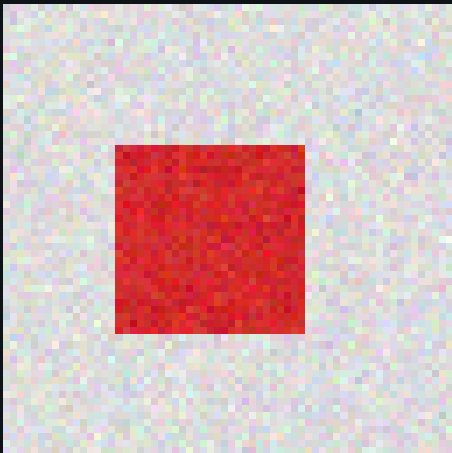
Five concepts determine the eight colour, shape and size classes.

Measured performance of the three architectures



Only the strict CBM sends the concept values into its class head.

Measured intervention: before



Test #10
Truth: small red square

red	1.000
blue	0.000
circle	0.039
square	0.988
large	0.782

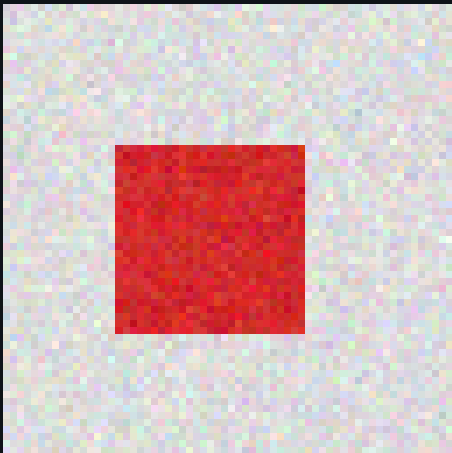
ACTUAL LEARNED HEAD

large red square

Same saved image
Same learned weights

The saved model predicts a large red square because of a wrong size estimate.

Measured intervention: after



Test #10
Truth: small red square

red	1.000
blue	0.000
circle	0.039
square	0.988
large	0.782 → 0

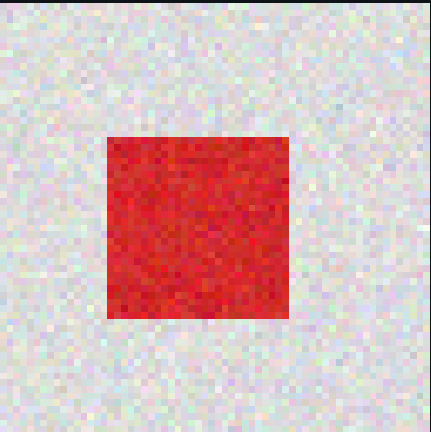
ACTUAL LEARNED HEAD

small red square

Same saved image
Same learned weights

Correcting large to zero changes the prediction to small red square.

Correct a concept in the trained shape model



Test #10
Truth: **small red square**

Same image and weights throughout the edit.

red	1.00
blue	0.00
circle	0.04
square	0.99
large	0.78

Saved predictions. Select one correction or move a slider.

In the HTML, move a slider to recompute.

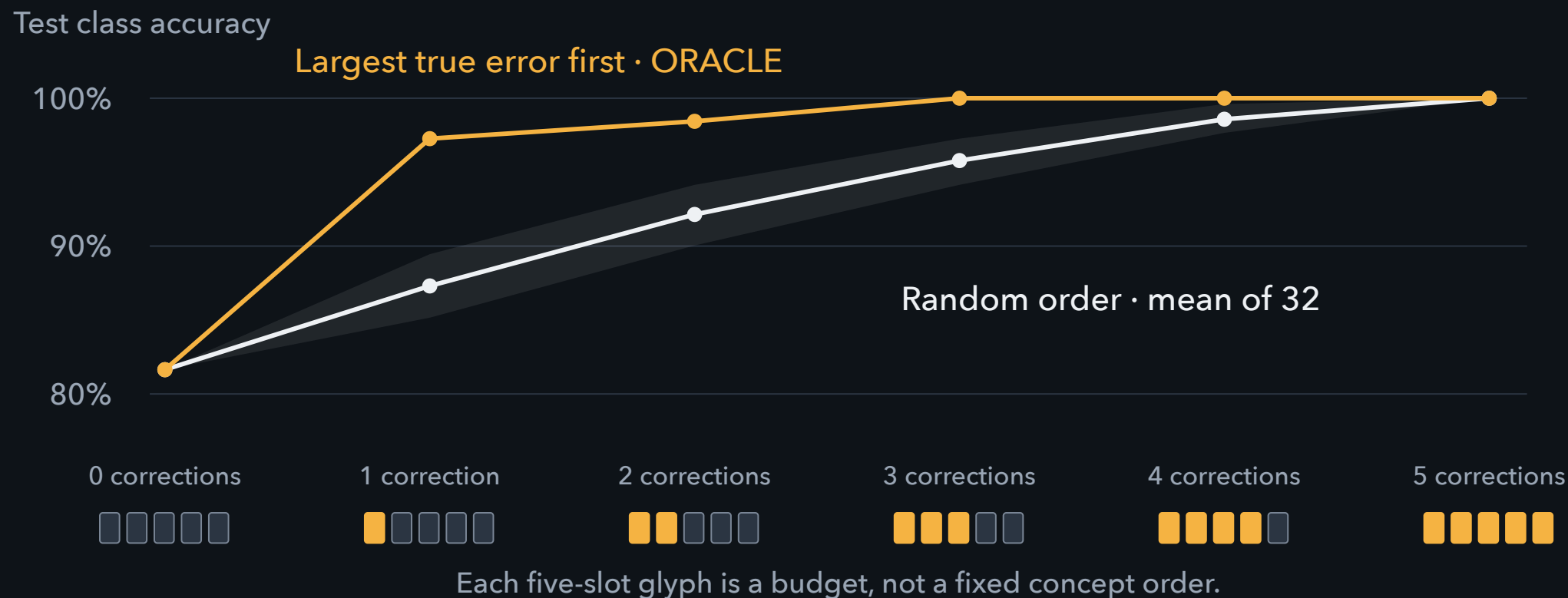
large red square

large red square	2.64
small red square	1.31
large red circle	0.57
large blue square	0.14

Numbers: learned class logits. Bars: softmax shares across 8 classes.

The sliders evaluate the learned task head on the current concept values.

Measured gains with ORACLE concept corrections



ORACLE: truth selects and corrects concepts. 81.6% → 97.3% after one correction.

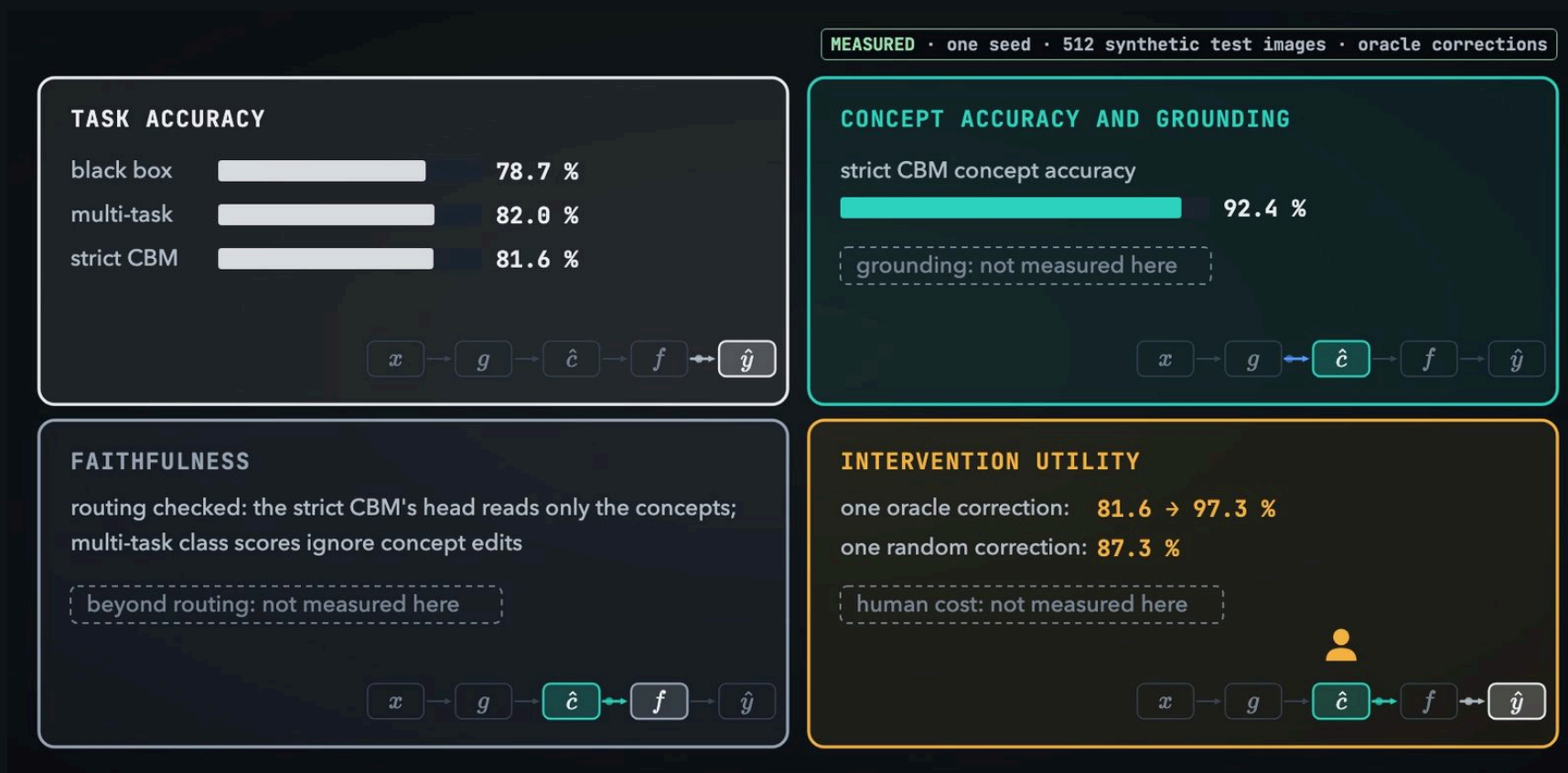
What did this experiment establish?

Task performance	Accuracy on 512 images	81.6%
Concept quality	2,560 binary labels	92.4%
Grounding faithfulness	Matched evidence / context tests	Not measured
Intervention utility	One oracle correction	97.3% task accuracy
Robustness under shift	New conditions or domains	Not measured

Technical measures and real-user usefulness require different evidence.

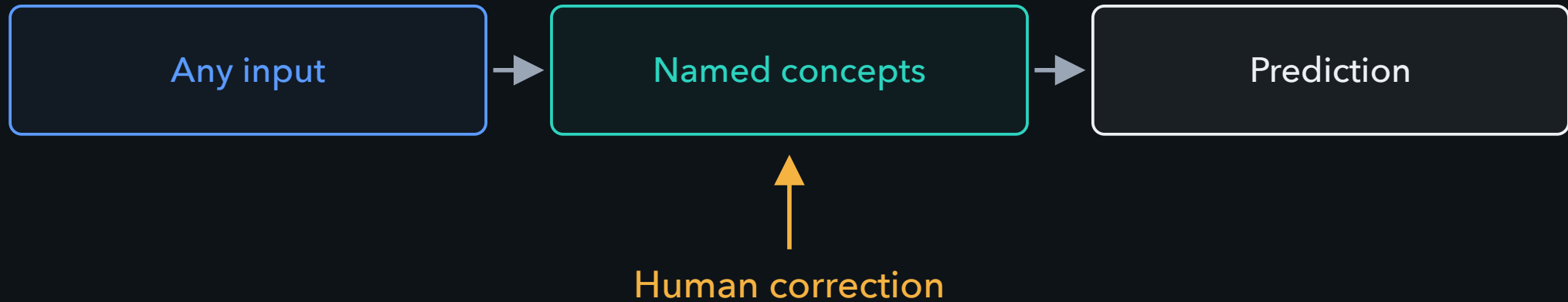
Task accuracy, concept accuracy and intervention performance answer different questions.

Four questions, measured and unmeasured



Task accuracy, concept quality, faithfulness and intervention utility are separate properties.

What to remember



Which variable should a human be able to inspect?

The five equations of a concept bottleneck

1 $\hat{\mathbf{c}} = g_{\theta}(x)$

Predict named concepts

input $\rightarrow$ concepts

2 $\hat{y} = f_{\phi}(\hat{\mathbf{c}})$

Predict through concepts

concepts $\rightarrow$ task

3 $\mathcal{L} = \lambda_c \mathcal{L}_c + \lambda_y \mathcal{L}_y$

Train both objectives

concept targets + task targets

4 $\hat{c}_j \leftarrow c_j^{(\text{human})}$

Correct a belief

human $\rightarrow$ one concept

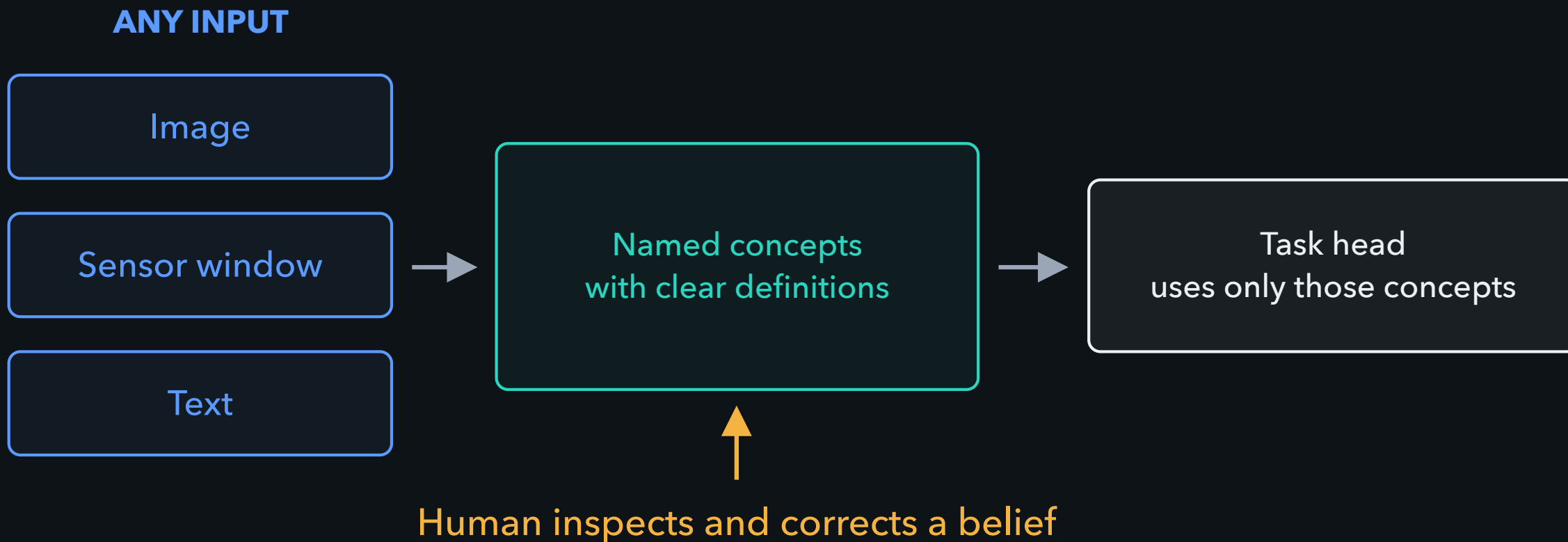
5 $\hat{y}_{\text{new}} = f_{\phi}(\hat{\mathbf{c}}_{\text{corrected}})$

Recompute the answer

edited concepts $\rightarrow$ same head

The correction replaces a concept value. The last equation reuses the same task head.

A concept bottleneck gives the human a place to intervene



An explanation tells us what the model may have used.
An intervenable bottleneck gives us something we can change.

Replay: the interface and its dangers

ILLUSTRATIVE VALUES · hand-written concept values and head



The challenge is to make sure the interface faithfully represents the information on which the prediction depends.

The challenge is to make the interface faithfully represent the information the prediction depends on.

Before you call a concept model interpretable

1. Who defines the concepts?
2. Are they observable and unambiguous?
3. How are they grounded?
4. Do they contain enough task information?
5. Can unintended information reach the head?
6. What exactly does an intervention change?
7. How will fidelity and usefulness be tested?

Define the interface. Test its meaning. Measure what happens when someone edits it.

Every stage can fail



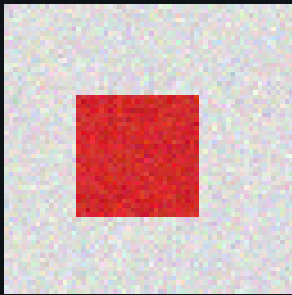
FIVE QUESTIONS FOR ANY CBM

- ✓ Are the concepts meaningful?
- ✓ Are they grounded correctly?
- ✓ Are they sufficient for the task?
- ✓ Does the model actually use them?
- ✓ Do corrections improve decisions?

Meaningful, grounded, sufficient, used, and helpful when corrected.

A selected success and aggregate evidence answer different questions

One example

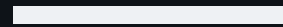


Test #10

large red square → small red square

All 29 one-bit-error cases

37.9% before



100% after



18 fixed · 0 harmed

Correct the one wrong bit

Scope

1 seed

Complete synthetic concepts

Perfect feedback

No real human trial

The measured result supports this mechanism under this experiment's assumptions.

An orientation, not a paper survey

2018

TCAV

User-defined concept probes

2019

ACE

Discovered visual explanations

2020

CBM

Supervised editable bottlenecks

2022

CEM

Richer concept representations

2023

PCBM · LF-CBM · LaBo

Pretrained features, VLMs and language

2023

Intervention-policy analyses

How feedback is selected and evaluated

Each method answers a different design or evaluation problem.

Primary sources behind the lecture

[Kim et al. · TCAV · ICML 2018](#)

[Koh et al. · Concept Bottleneck Models · ICML 2020](#)

[Yuksekgonul et al. · Post-hoc CBMs · ICLR 2023](#)

[Yang et al. · Language in a Bottle · CVPR 2023](#)

[Radford et al. · CLIP · ICML 2021](#)

[Ghorbani et al. · ACE · NeurIPS 2019](#)

[Zarlenga et al. · Concept Embedding Models · NeurIPS 2022](#)

[Oikarinen et al. · Label-Free CBMs · ICLR 2023](#)

[Shin et al. · A Closer Look at the Intervention Procedure · ICML 2023](#)

[Mahinpei et al. · Promises and Pitfalls of Black-Box Concept Learning · 2021](#)

Each method solves a specific problem; no single axis ranks them all.

One edit can inform other concept estimates

SCALAR REPLACEMENT

Correct one selected concept



Other concept estimates
stay fixed

STOCHASTIC CONCEPT MODEL

Correct one selected concept



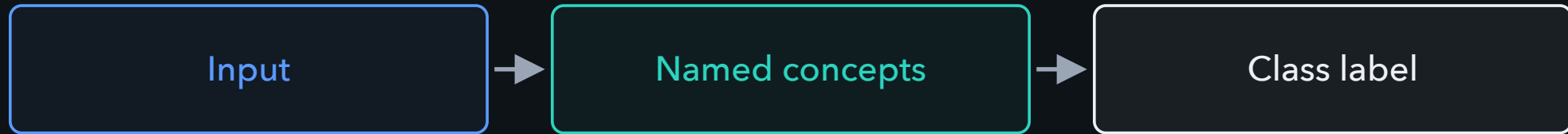
Update beliefs about
correlated concepts

A statistical dependency is not, by itself, a causal relationship.

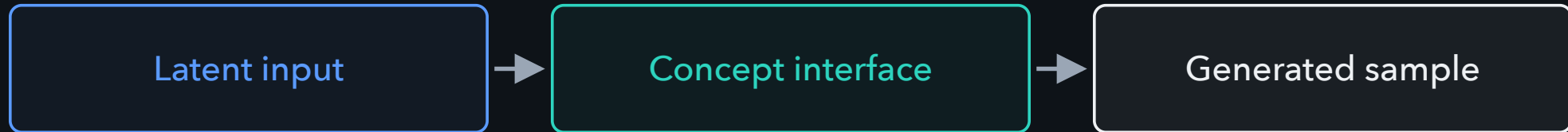
Stochastic CBMs use a joint concept model to propagate information from an edit.

Concept interfaces can also steer generation

CLASSIFICATION



GENERATION



▲
Edit a concept to steer a property of the generated sample

Audit any additional representation or conditioning path.

The output can be a generated sample rather than a class label.

Read next: design, grounding and reliable edits

[Knab et al. · What's in the Bottle? · TMLR 2026](#)

A modular taxonomy and a research roadmap.

[Vandenhirtz et al. · Stochastic CBMs · NeurIPS 2024](#)

Use dependencies between concepts during correction.

[Margeloiu et al. · Do CBMs Learn as Intended? · 2021](#)

Do the learned concepts match their intended input meaning?

[Ismail et al. · Concept Bottleneck Generative Models · ICLR 2024](#)

Expose concept controls within generative models.

Start with a clear definition, inspect the information path, then test the edits.

The three training strategies together

INDEPENDENT

$$x \rightarrow g_{\theta} \rightarrow \hat{c}$$

$$c \rightarrow f_{\phi} \rightarrow \hat{y}$$

Task head fits true concepts

Two separately fitted models

SEQUENTIAL

$$x \rightarrow g_{\theta} \rightarrow \hat{c}$$

FREEZE concept predictor

$$\hat{c} \rightarrow f_{\phi} \rightarrow \hat{y}$$

Task head fits predictions

JOINT

$$x \rightarrow \hat{c} \rightarrow \hat{y}$$

$\mathcal{L}_c$

$\mathcal{L}_y$

Both losses train together

Deployment: every strategy receives predicted concepts.

The strategies differ in what the task head sees during training.

Reference: the exact illustrative task head

CONCEPT ORDER

stripes · horse shape · long neck

$$W = \begin{bmatrix} 3 & 1 & 0 \\ -1 & 2 & -1 \\ -1 & 0 & 3 \end{bmatrix}$$

Rows: zebra, horse, giraffe

$$\hat{\mathbf{c}} = \begin{bmatrix} 0.1 \\ 0.9 \\ 0.1 \end{bmatrix}$$

Three predicted concepts

W: 3 × 3 concepts: 3 × 1 logits: 3 × 1 bias: zero

Use this W for the arithmetic and the interactive toy demonstration.

Six places a concept-based system can fail

Estimate

The detector misreads the input.

Size: large instead of small.

Vocabulary

A needed distinction is missing.

Colour and size omit shape.

Task head

The mapping from concepts is wrong.

Negative duplicate-charge weight.

Meaning / evidence

The concept follows a shortcut.

Background instead of chimney.

Leakage

Scores carry undeclared information.

0.72 versus 0.96 encodes class.

Feedback

The human supplies a wrong value.

Stripes marked present on a horse.

Identify the failing component before choosing a repair.

Suggested tests across images, signals and text

CHANGE THE CONDITIONS

Image: background or viewpoint

EEG: sensor or recording setting

Text: negation or phrasing

REPEAT THE CHECKS

→ Concept accuracy + masking controls

→ Event grounding + unknown cases

→ Meaning consistency + routing effect

Hold the task definition fixed. Report harms, gains and correction effort.

A result under one set of conditions does not establish robustness elsewhere.

Sources for signals and text

[Jeyakumar et al. · X-CHAR · 2023](#)

Concepts for human activity recognition from sensor data.

[Lucey et al. · EEG sleep staging study · 2016](#)

Sleep-scoring context for spindle and K-complex examples.

[Ludan et al. · Text Bottleneck Models · 2024](#)

A text classifier that predicts through concept values.

Our waveforms, message and example scores are authored teaching cases.